

Day 4 · Generative Audio & Video

Sound, Motion, Story.

How generative AI creates voices, music, sound effects, movement, and video—and how people turn them into meaningful stories.



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Three sections · ten modules

What we'll explore

We begin with voice and audio models, move into video generation, and finish by comparing how these systems learn, generate, and work together.

Section 1

Generating sound

- 01 Voice as model input
- 02 Generating speech
- 03 Music & sound effects

Section 2

Generating video

- 04 Video as a sequence
- 05 Noise to video
- 06 Temporal coherence
- 07 Directing generation

Section 3

Model systems

- 08 Multimodal generation
- 09 Training & inference
- 10 Architectures & limits

Generative systems predict patterns in different media

The representation changes, but the central idea remains: context guides a sequence of small predictions.

Language and images

Tokens and visual change

Words continue across a sequence; images emerge through repeated denoising.

text · pixels · latent representations

Audio and video

Patterns unfold through time

Sound samples and video frames must remain connected from one moment to the next.

waveform · frames · motion

A short bridge from voice to models

Voice as model input

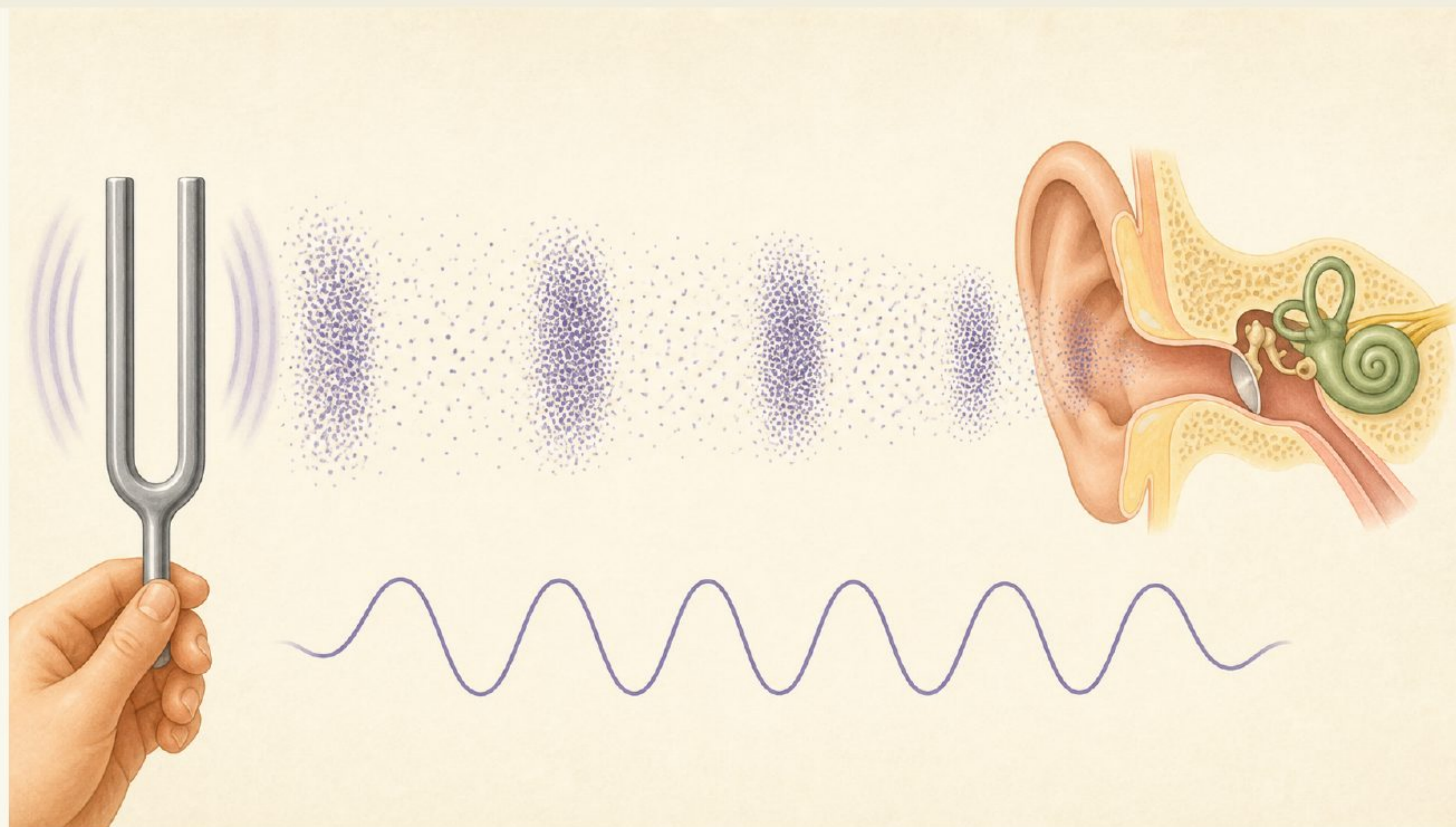
We only need one foundation: a microphone turns a voice into numbers, then learned encoders convert those numbers into representations a model can predict.



Vibration becomes a measurable signal

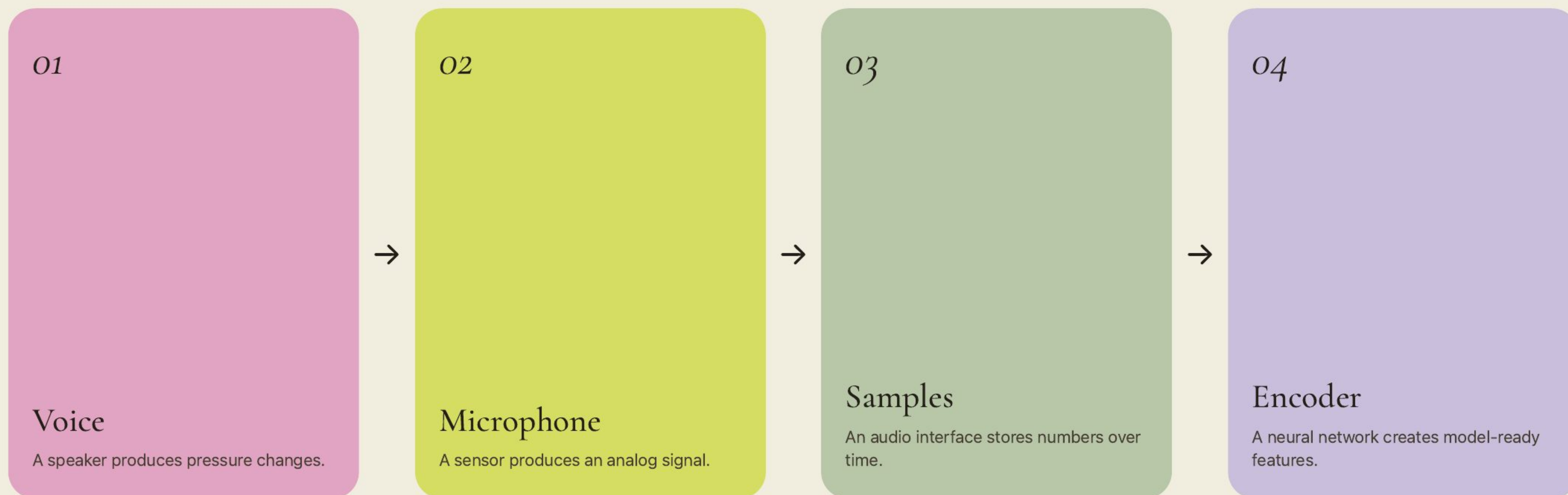
Voice begins as vibrations moving through air

Our vocal cords create tiny pressure changes. A microphone measures that pattern and stores it as digital samples. From this point on, the model works with numbers.



Voice reaches the model through four conversions

Each stage changes the format while trying to preserve the linguistic and acoustic information people hear.



A voice signal mixes several factors at once

A generation model must preserve the words while also deciding timing, speaker traits, and recording conditions.

01

Content

Phonemes and words carry the message.

02

Prosody

Timing, stress, pitch, and pauses shape delivery.

03

Speaker

Timbre and vocal habits help define identity.

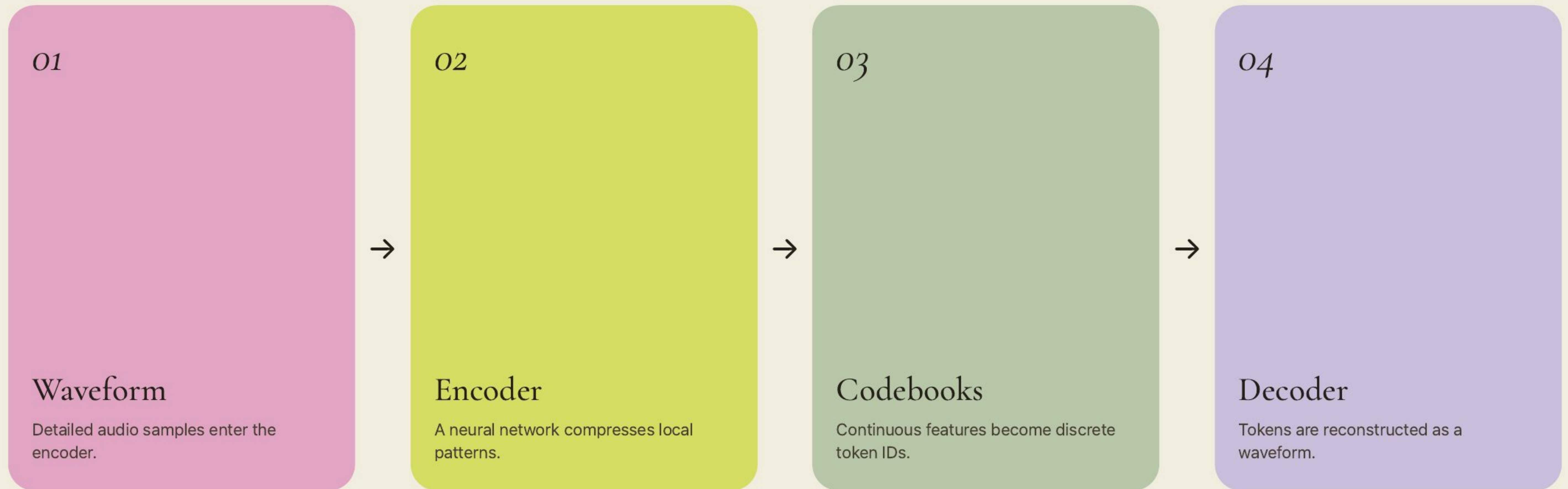
04

Acoustics

Microphone, room, noise, and distance affect the signal.

An encoder can turn audio into compact tokens

A learned codec compresses the waveform, quantizes the hidden representation, and later reconstructs audible sound.



The representation changes what the generator predicts

A model can predict tiny sample values, a smooth latent, or token IDs selected from a learned codebook.

Continuous

Waveform or latent values

Precise numerical values preserve detail but may require many predictions or denoising steps.

samples · spectrogram · latent

Discrete

Neural codec tokens

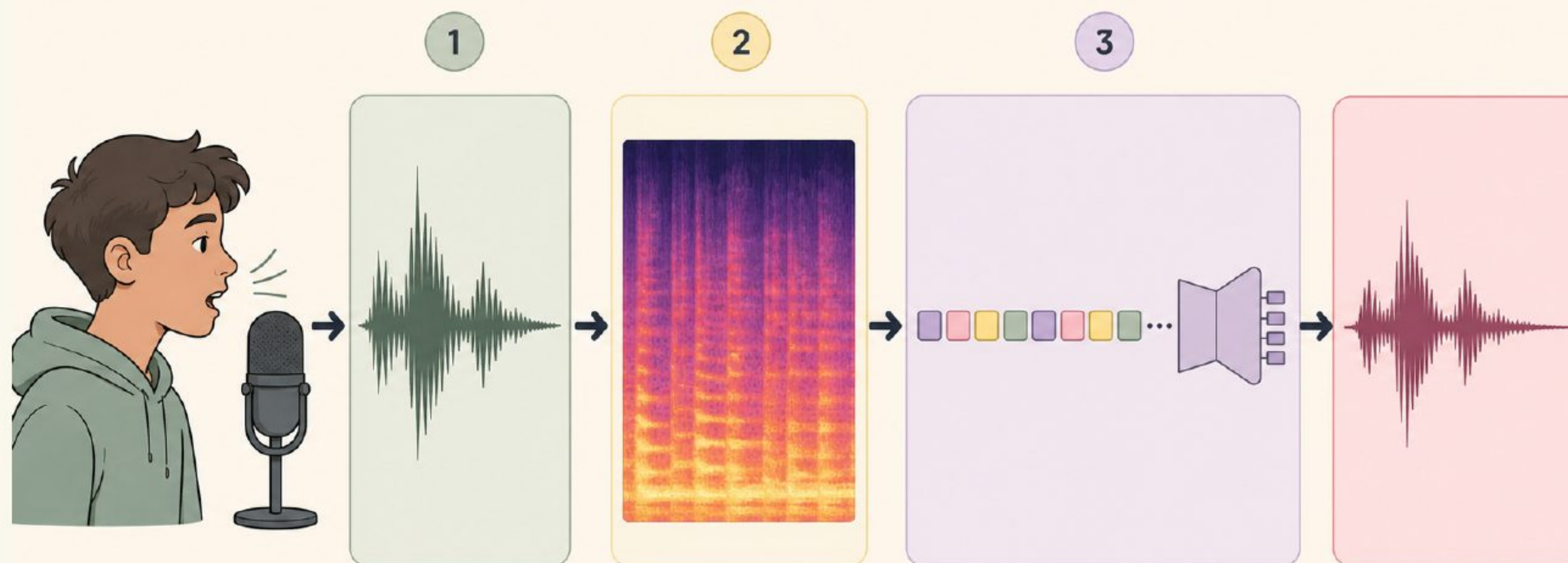
A transformer can treat compressed audio units like a sequence of learned symbols.

token 418 → token 92 → token 701

Waveform → spectrogram → codec tokens

Models choose a representation before they generate

The waveform keeps sample detail, the spectrogram exposes frequency structure, and codec tokens compress the sound into learnable units.



A model can work with sound at different levels

The best representation depends on whether the system needs precise samples, visible frequency structure, or compact learnable units.

01

Waveform

A long sequence of sample values. It keeps fine detail, but is expensive to generate directly.

02

Spectrogram

A time–frequency map. It makes pitch, harmonics, and changing energy easier to model.

03

Codec tokens

A neural codec compresses audio into discrete units that a sequence model can predict.

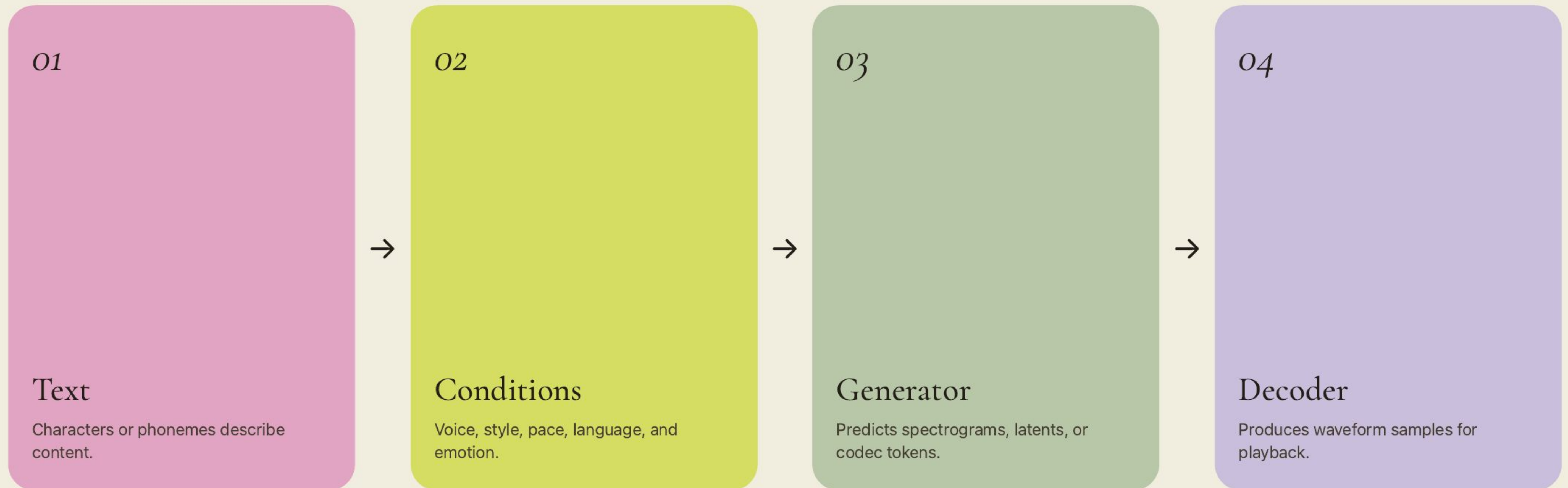
Words must become timing and sound

Generating speech

Text-to-speech systems learn how written language relates to pronunciation, rhythm, pitch, expression, and audio.

A TTS model predicts acoustics from text and conditions

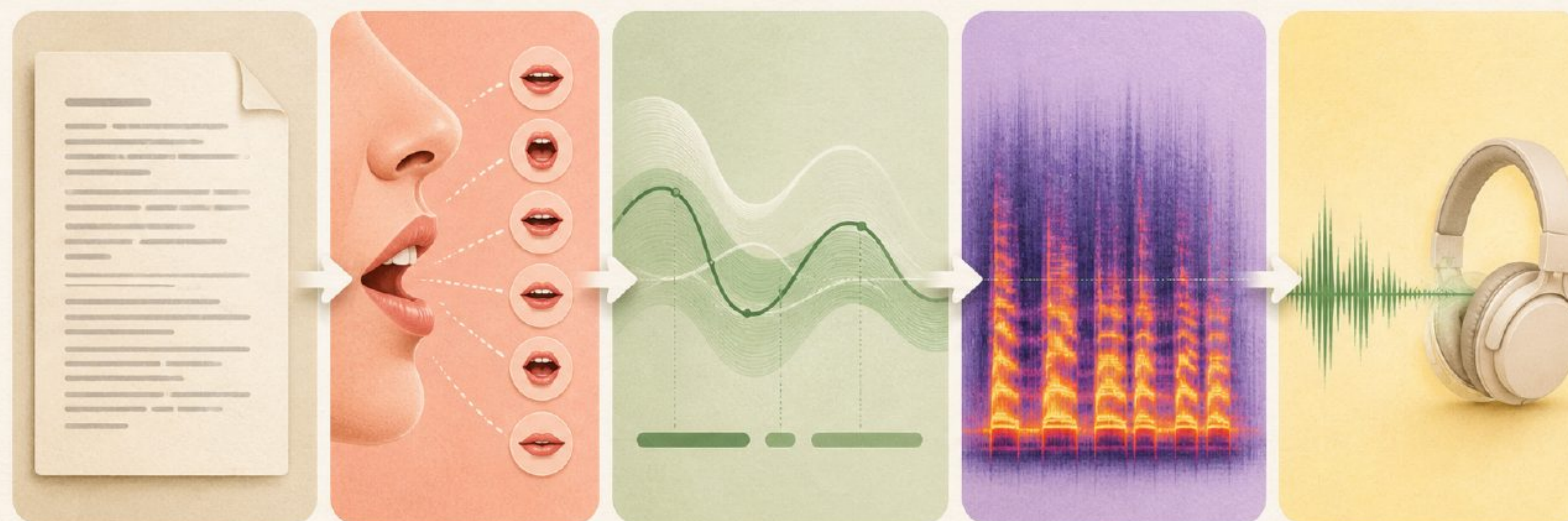
Modern systems separate what should be said from how it should sound, then synthesize an audio representation before decoding the waveform.



Five connected representations

Speech generation builds several layers

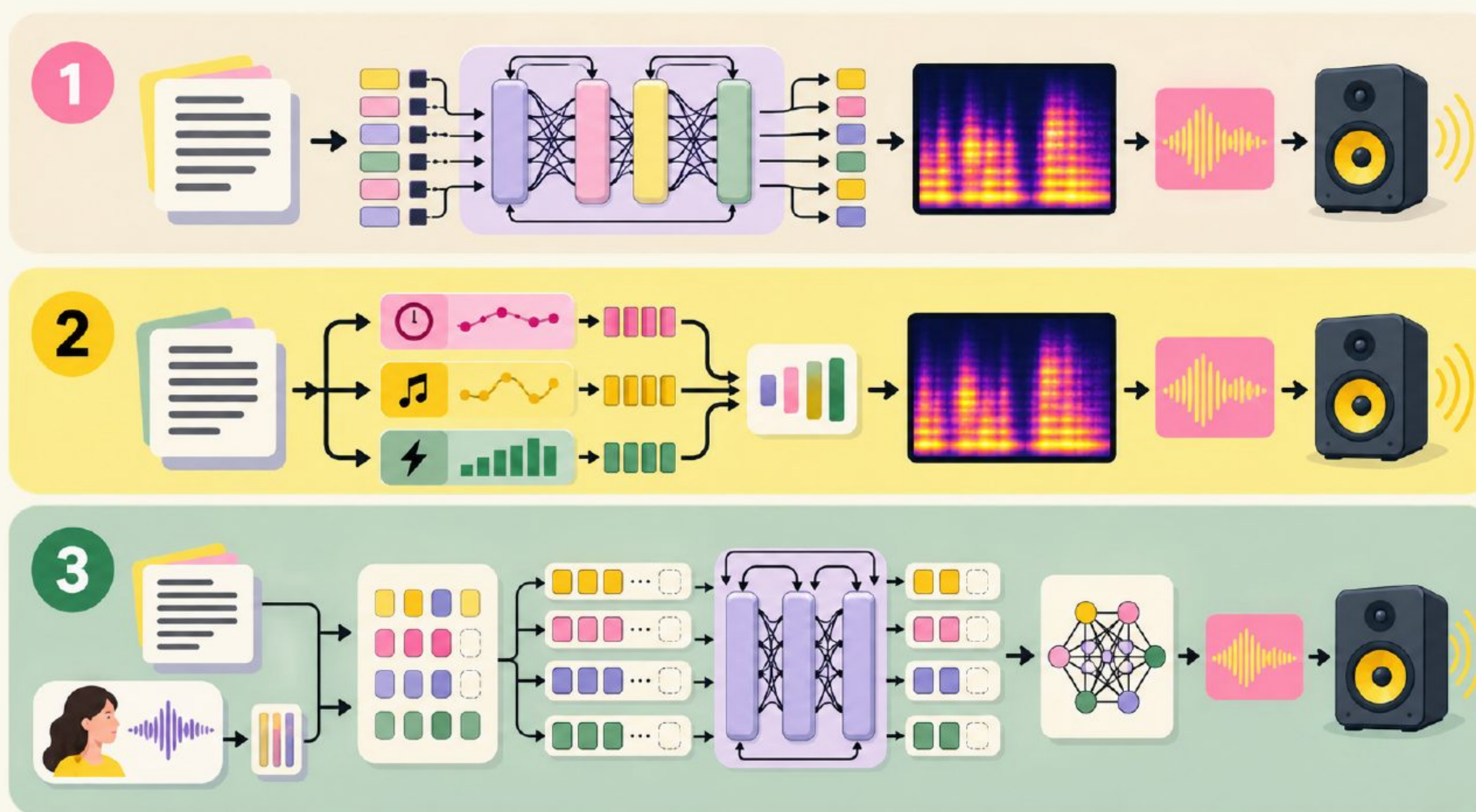
Text becomes pronunciation and prosody, then an audio representation, and finally a waveform we can hear.



Different internal paths · the same audible goal

Three TTS architectures organize prediction differently

Tacotron predicts spectrogram frames in sequence, FastSpeech predicts controllable features in parallel, and codec language models predict compressed speech tokens.



Generation order changes speed and control

The architecture decides whether speech frames depend on earlier generated frames or can be predicted together.

Tacotron-style

Predict the next acoustic frame

Attention connects text positions to a growing spectrogram. Earlier output becomes context for the next step.

accurate alignment · slower sequential decoding

FastSpeech-style

Predict frames in parallel

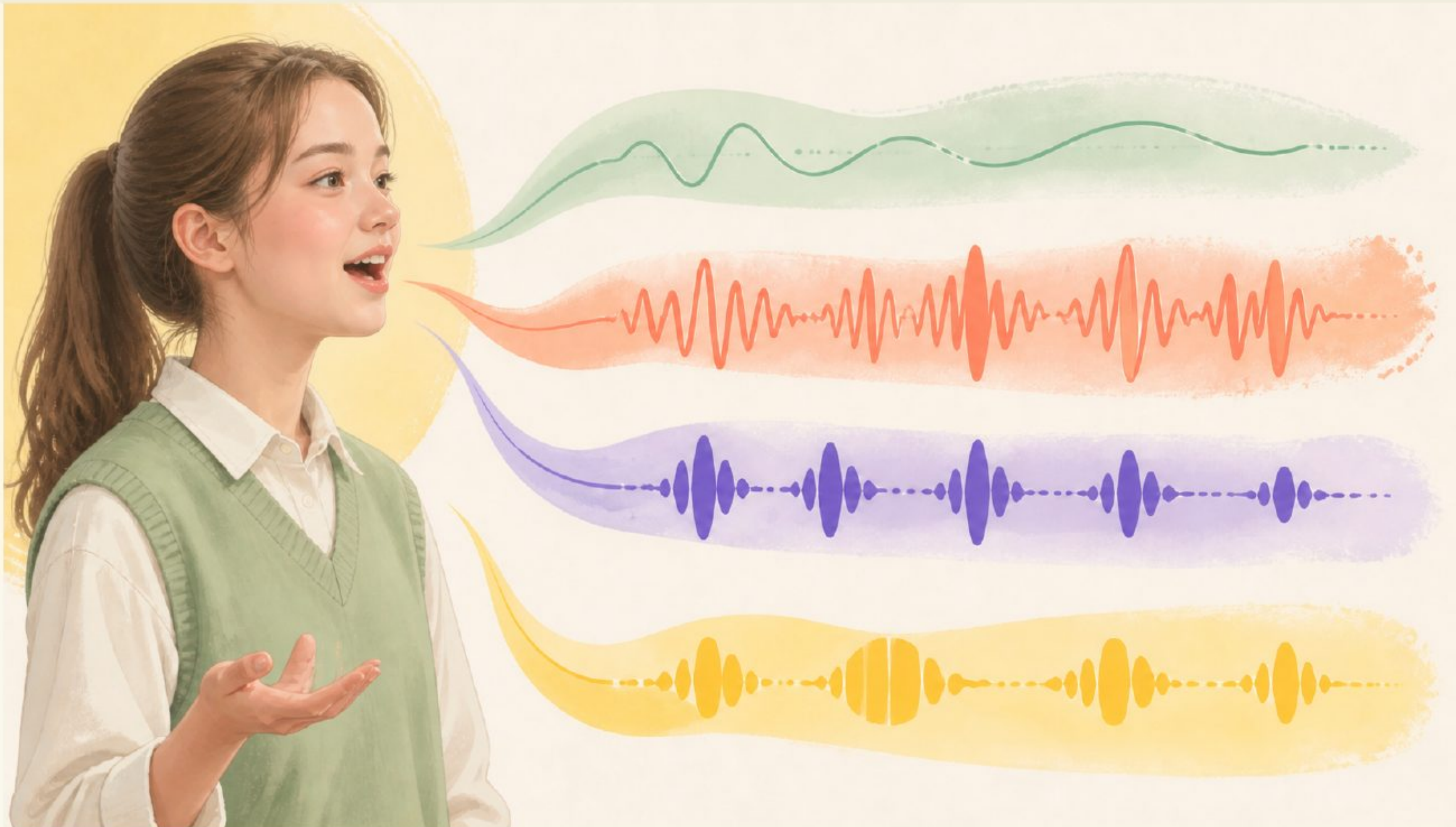
Duration, pitch, and energy predictors expand text features into a full acoustic sequence at once.

faster synthesis · explicit controls

Direct the performance

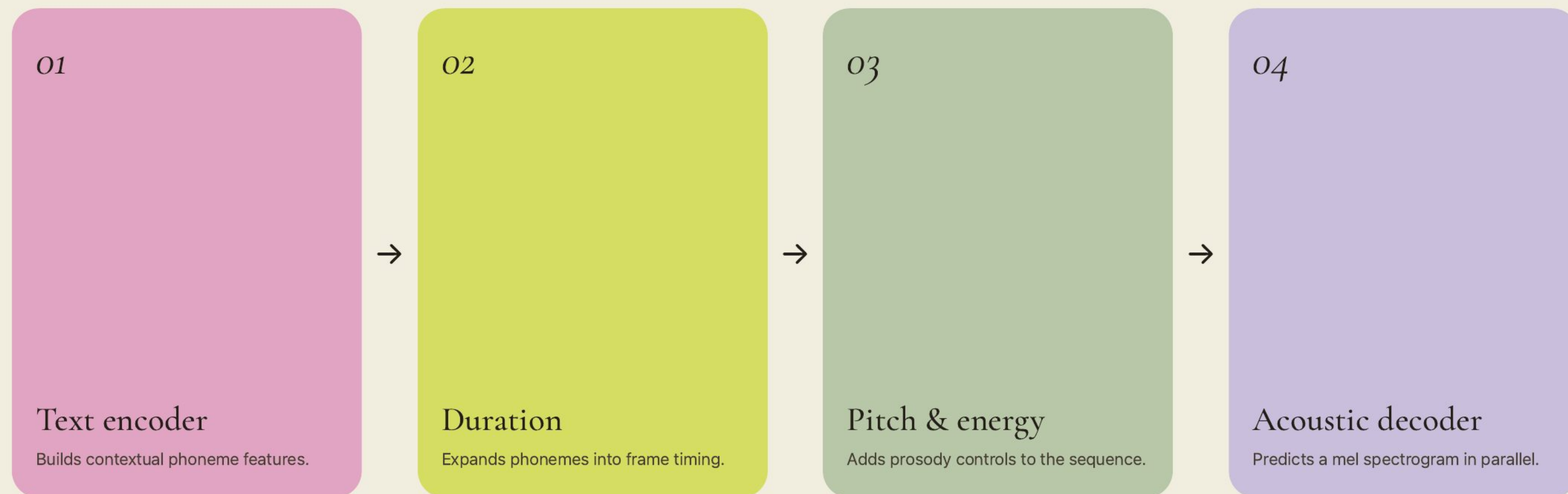
The same words can sound calm, excited, urgent, or warm

The abstract ribbons show differences in pace, intensity, pitch movement, and pauses.



Variance predictors make speech controllable

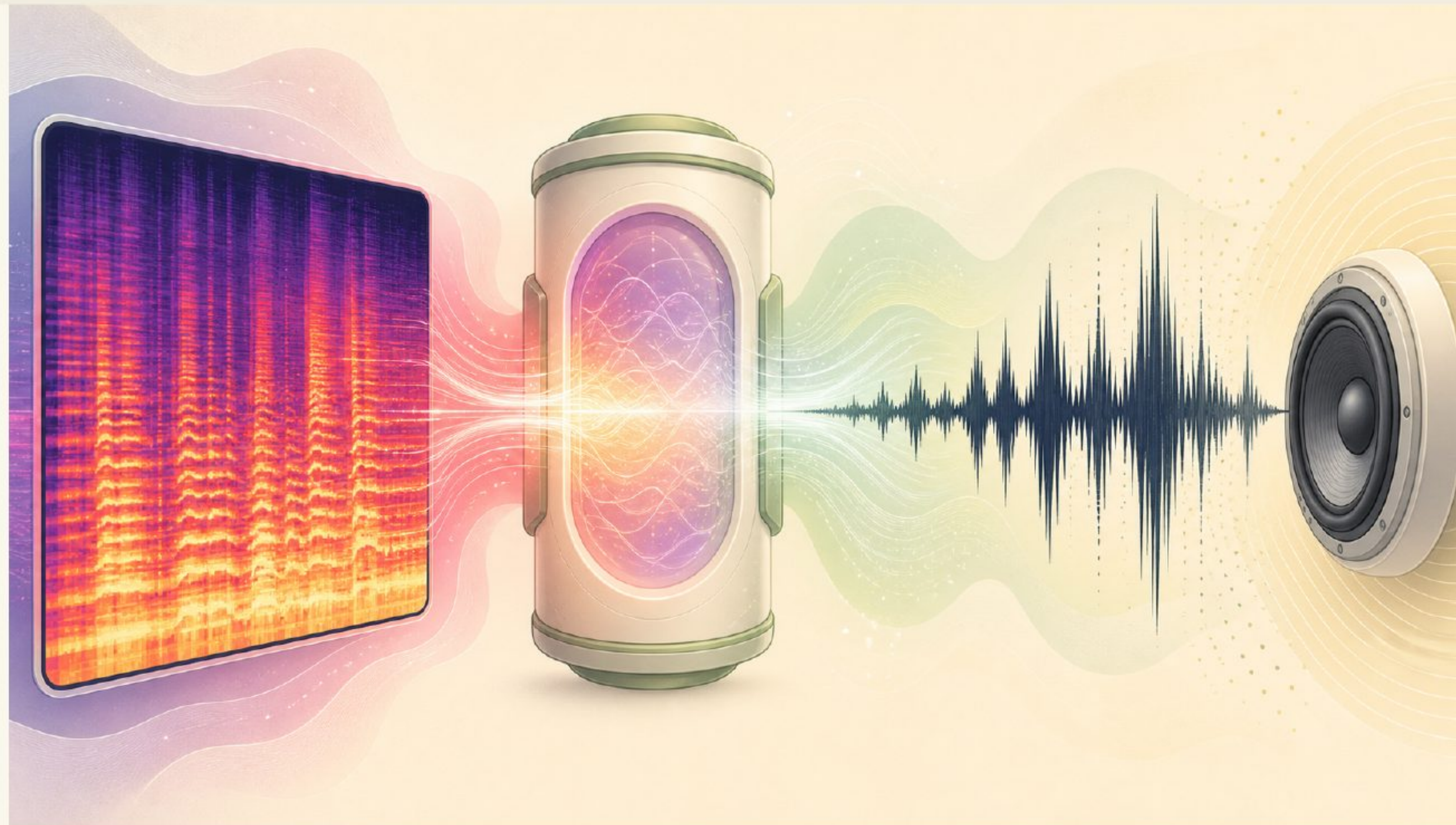
The model combines text features with predicted duration, pitch, and energy before producing the acoustic representation.



Representation becomes audio

A vocoder reconstructs fine audio detail

The spectrogram describes changing frequency energy. The vocoder turns that description into waveform samples for a speaker.



The representation defines the prediction task

All three families begin with language and end with a waveform. What changes is the structure predicted in the middle—and whether it is generated sequentially, in parallel, or as tokens.

01

Tacotron-style

Predict the next mel-spectrogram frame from text, attention, and previously generated acoustic frames.

02

FastSpeech-style

Predict duration, pitch, energy, and many mel-spectrogram frames in parallel for speed and direct control.

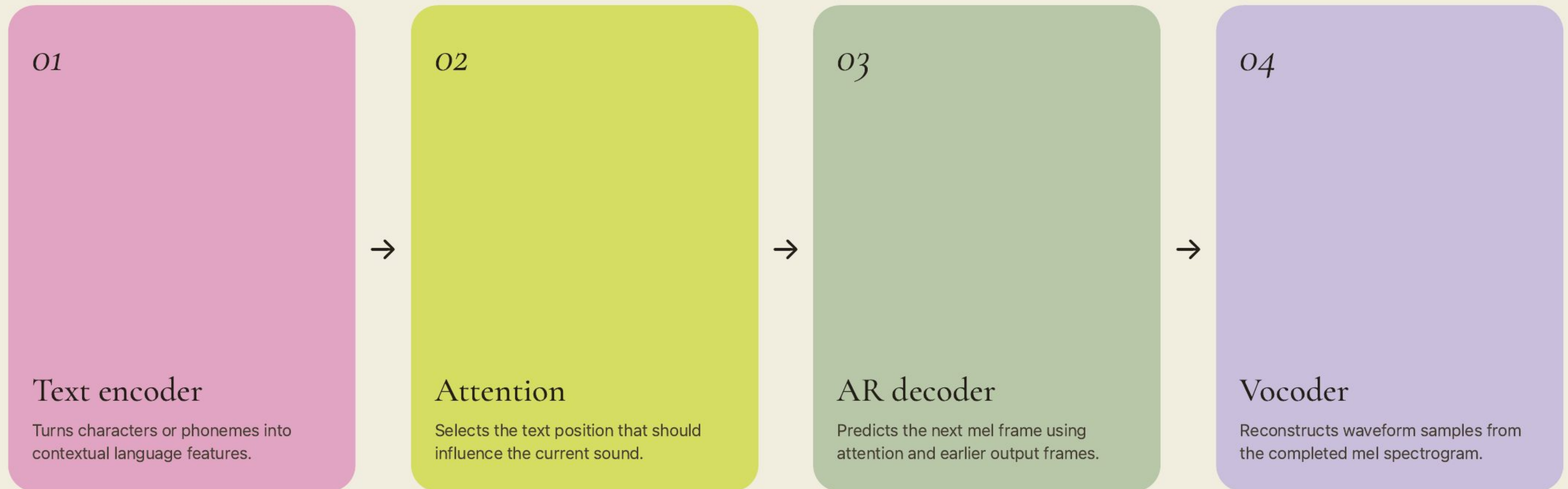
03

Codec language model

Compress speech into discrete audio tokens, predict them like language, and decode the completed token stream.

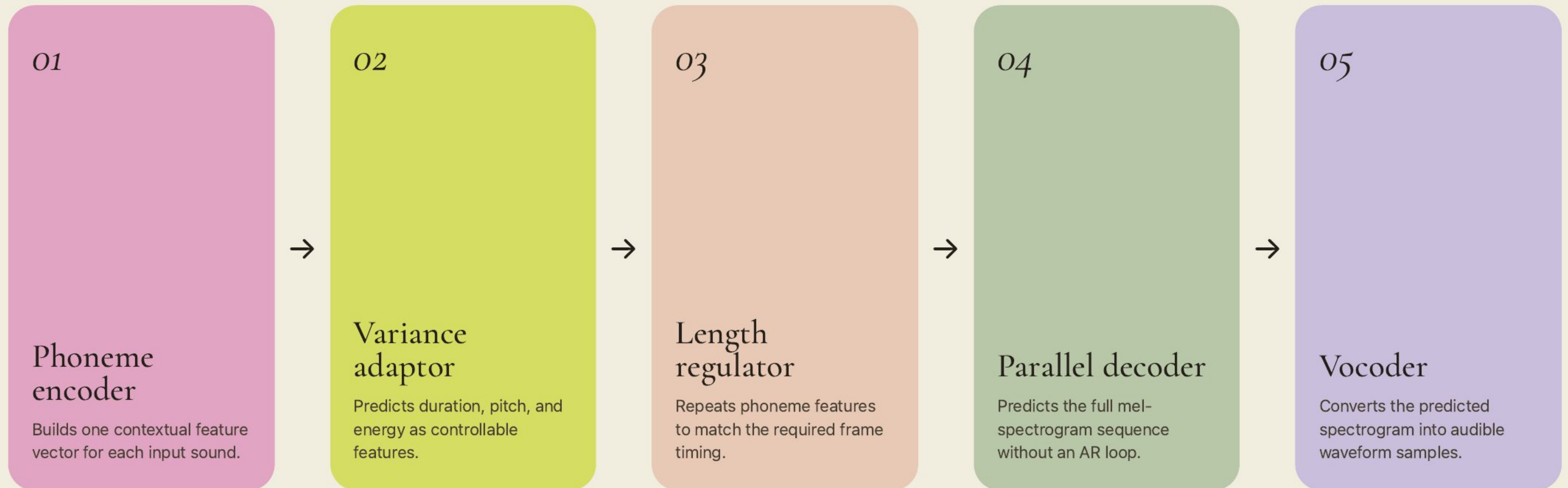
Attention aligns text with one acoustic frame at a time

Tacotron is autoregressive: each new spectrogram frame depends partly on frames already generated. This creates a flexible text-to-audio alignment, but sequential decoding can be slow and errors may accumulate.



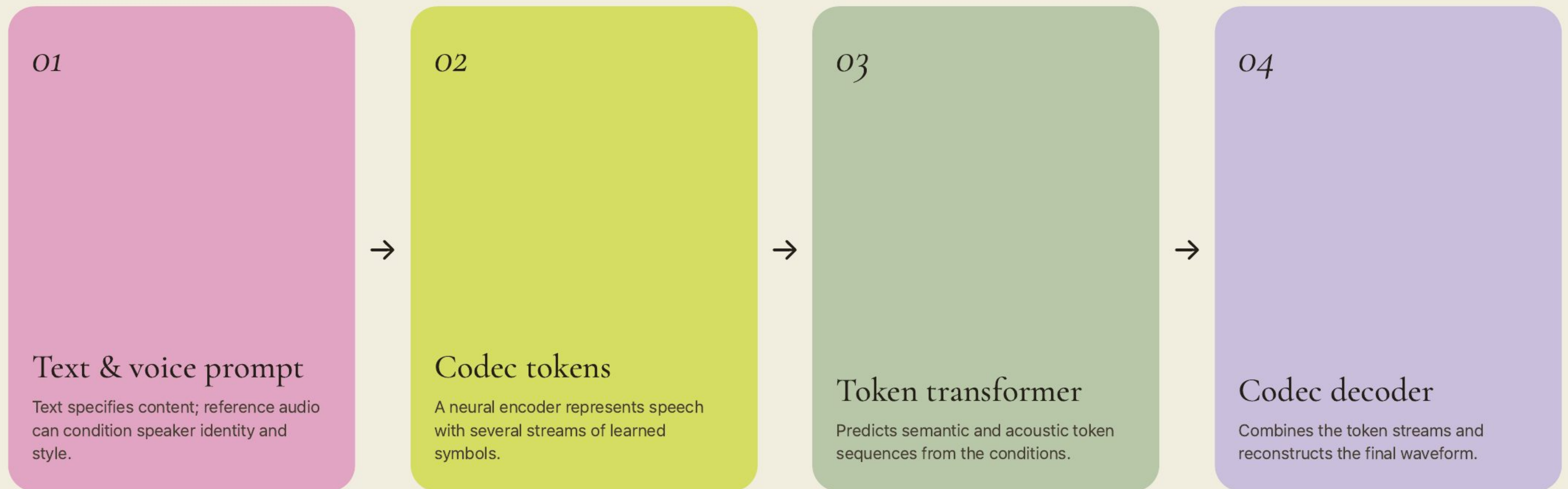
Explicit timing controls enable parallel prediction

FastSpeech removes the frame-by-frame feedback loop. A variance adaptor predicts how long, high, and energetic each sound should be, then expands the phoneme sequence before the acoustic decoder works in parallel.



Speech can be generated as a hierarchy of audio tokens

A neural codec replaces dense waveform samples with compact discrete symbols. A transformer predicts these symbols using text and an optional voice prompt, then the codec decoder reconstructs speech.



Speech is only one kind of audio

Music & sound effects

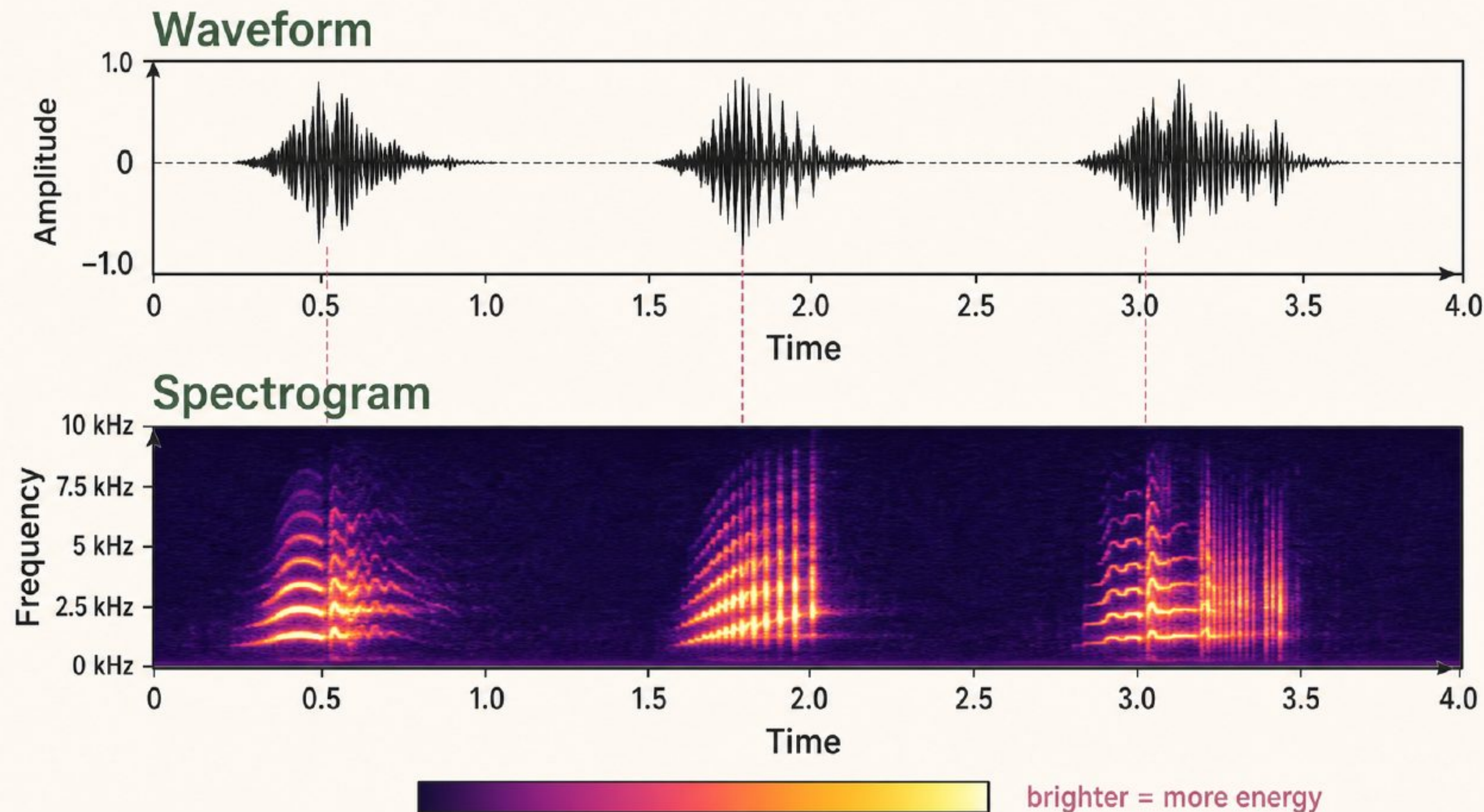
Generative audio models learn patterns in rhythm, pitch, texture, structure, and everyday acoustic events.



The horizontal axis is time in both diagrams

Waveforms show amplitude; spectrograms show frequency

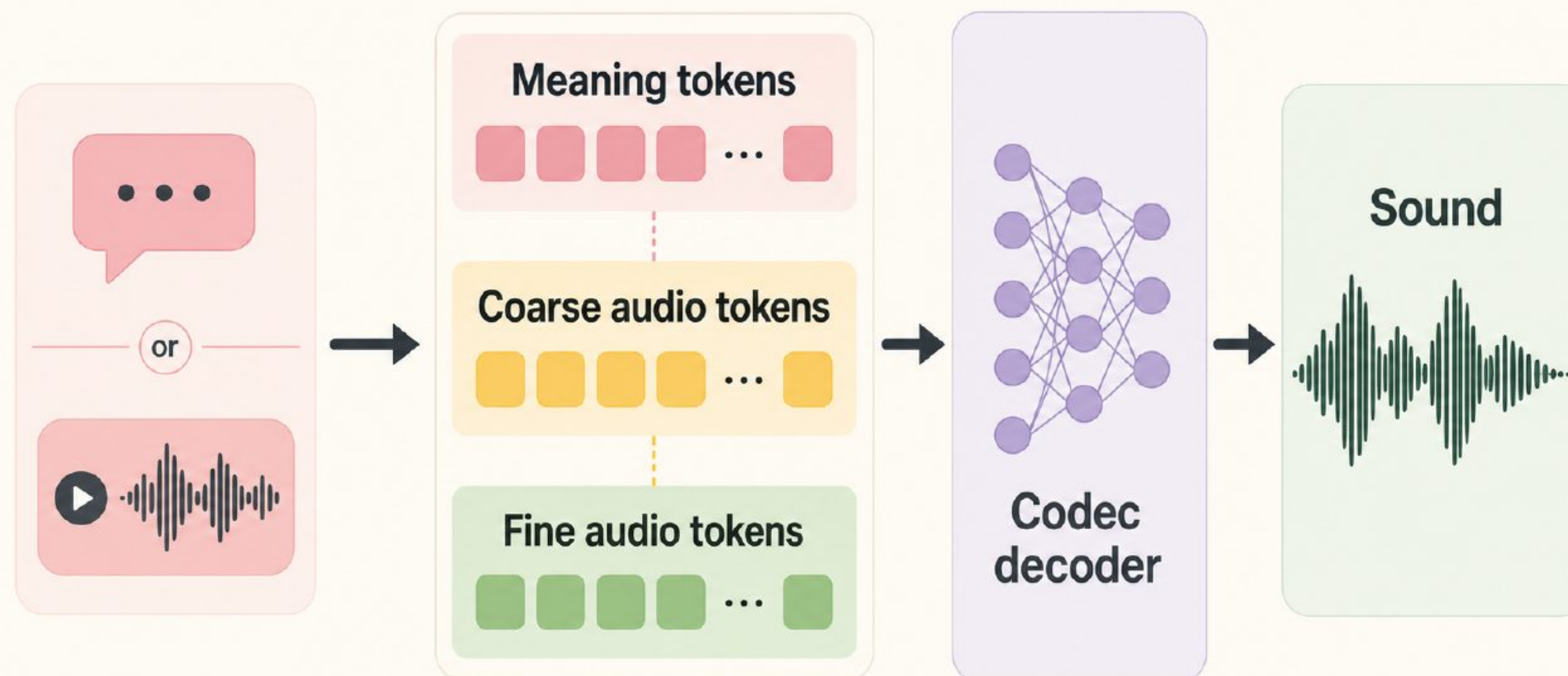
A waveform shows how signal amplitude changes over time. A spectrogram shows which frequencies are active at each time, while brighter color means stronger energy.



Plan meaning first · add sound detail later

AudioLM predicts sound at several levels

The first tokens describe what is happening. Later token layers add speaker, instrument, texture, and fine acoustic detail before the codec reconstructs sound.



The generator can receive more than a text prompt

Different model families combine conditions that control content, style, structure, and timing.

01

Text embedding

Describes events, instruments, mood, or scene.

02

Reference audio

Provides voice, melody, rhythm, or acoustic style.

03

Timing controls

Specify duration, tempo, or event placement.

04

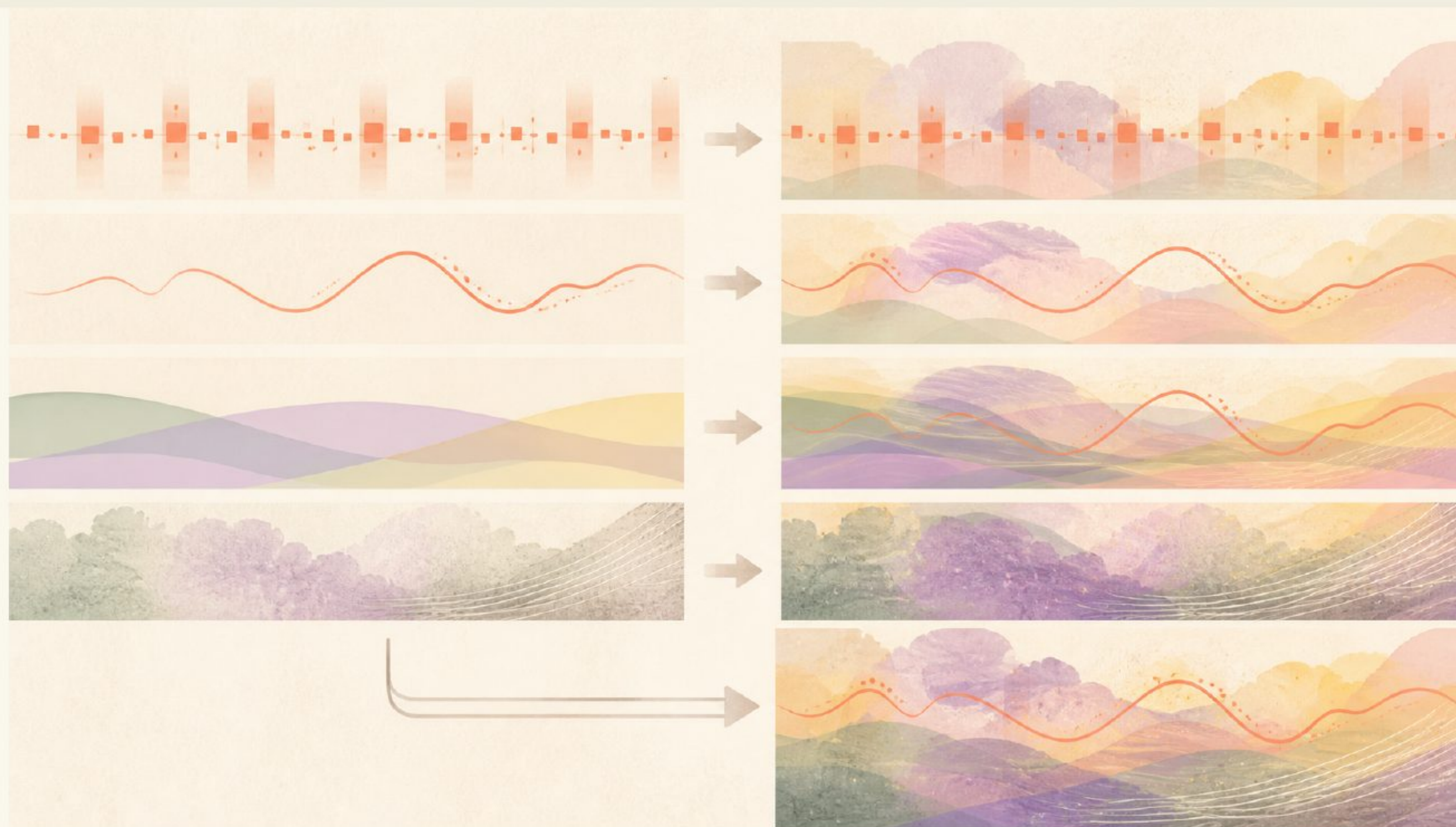
Audio context

A prefix gives the model material to continue.

Separate ingredients · one experience

Pulse, melody, harmony, and texture combine

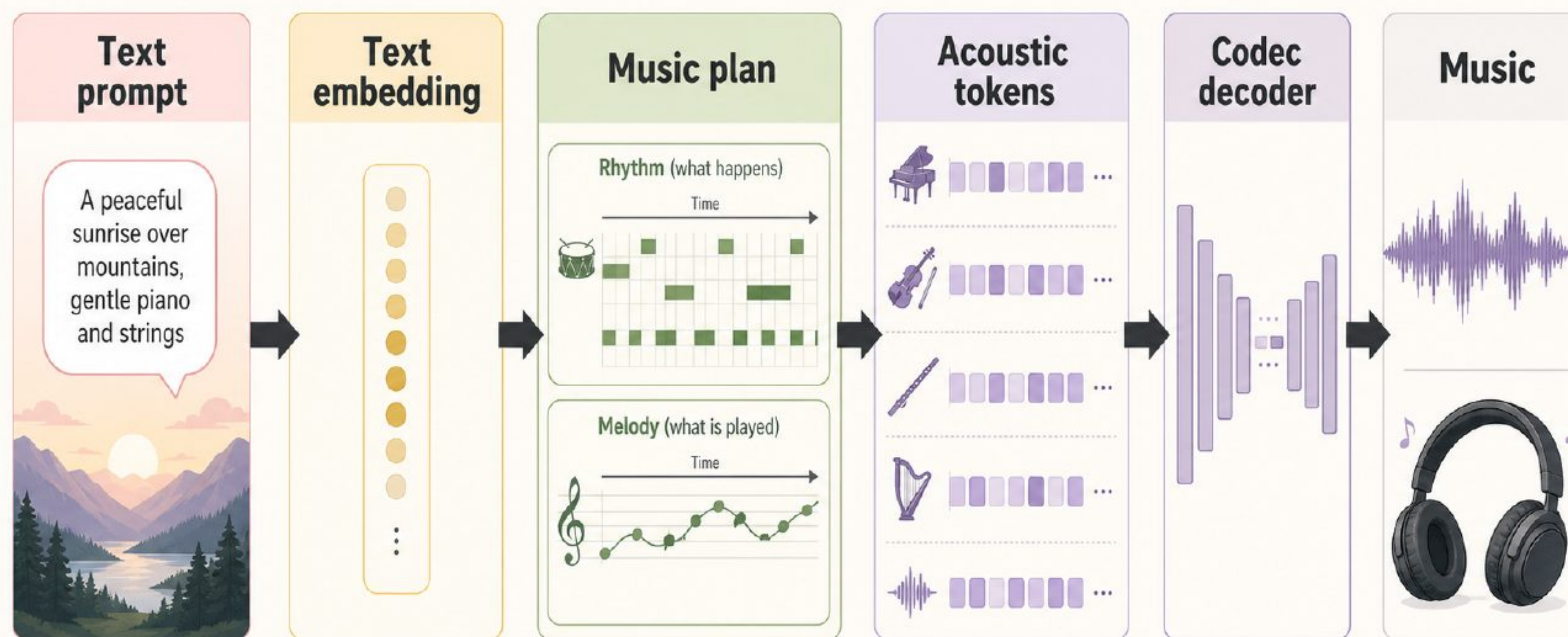
Each visual band follows its own pattern while contributing to the overall direction of the music.



Language guides a musical plan

MusicLM separates the plan from the sound texture

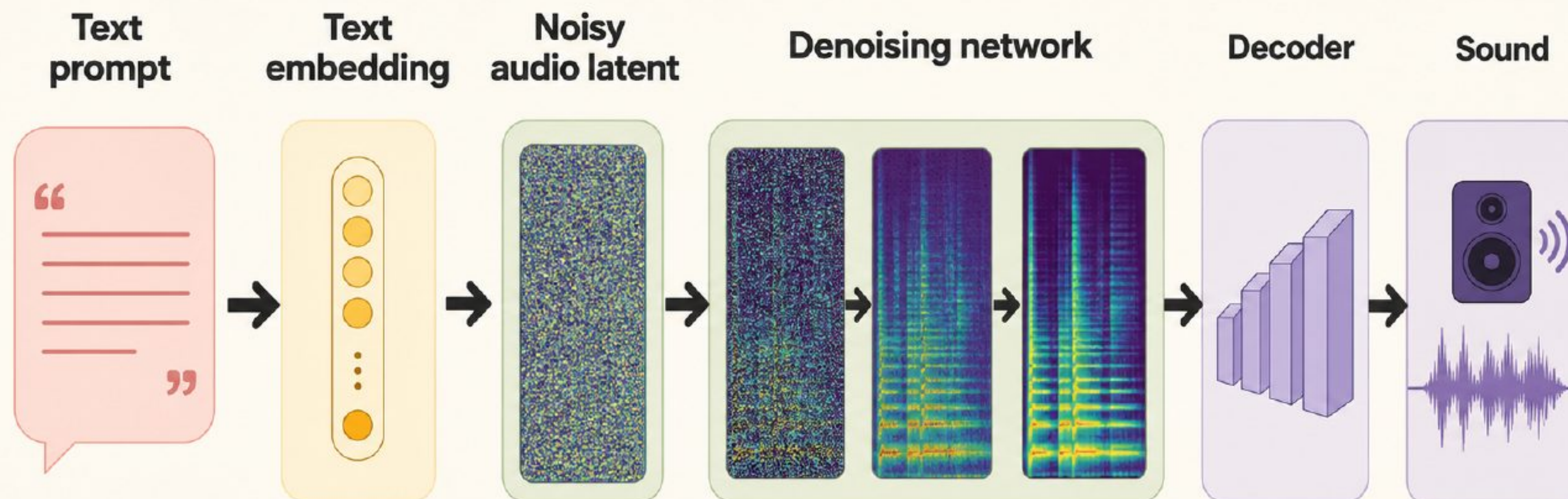
A text embedding is a learned list of numbers that represents the prompt. The model plans rhythm and melody, adds instrument-level audio tokens, and decodes them into music.



Diffusion works on a compressed sound picture

AudioLDM cleans a noisy audio latent step by step

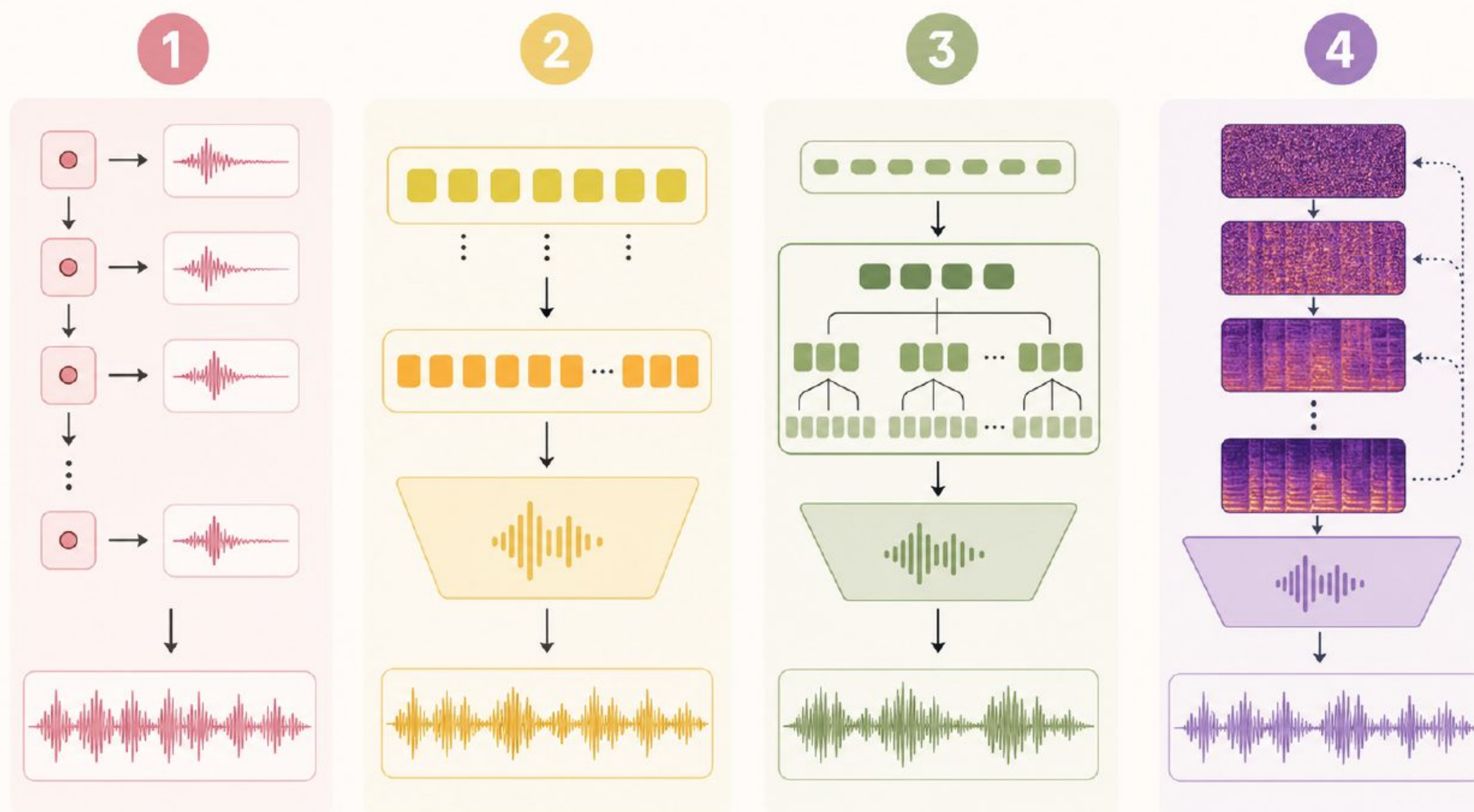
A latent is a smaller learned representation of sound. The model denoises that compact representation under text guidance, then a decoder reconstructs audible audio.



Samples · tokens · hierarchies · diffusion

Audio models differ mainly in what they predict

WaveNet predicts waveform samples, AudioLM and MusicLM predict token hierarchies, and AudioLDM denoises a compact latent representation.



Three model families turn conditions into sound

Modern systems often generate a compact audio representation first, then decode it into a waveform that we can hear.

01

Autoregressive tokens

Predict one audio token after another. AudioLM is a foundational example of this sequence approach.

02

Text-to-audio tokens

AudioGen conditions a token generator on text describing events, environments, and sound sources.

03

Latent diffusion

AudioLDM repeatedly denoises a compact audio representation, then reconstructs the final waveform.

Audio generation must coordinate time

Speech, music, and sound effects all depend on patterns that unfold rather than appearing all at once.

01

Represent

Sound becomes samples and learned audio representations.

02

Condition

Text guides voice, mood, structure, and events.

03

Listen

Human review checks clarity, timing, quality, and purpose.

A moving image is more than one picture

Video is a sequence

Video creates the experience of motion by presenting related frames across time.



Video adds time to every visual representation

A generator may process full frames, compressed latent blocks, space-time patches, or discrete video tokens.

01

Pixels

Full-resolution color values preserve detail but require heavy computation.

02

Latent blocks

A video autoencoder compresses space and time before generation.

03

Space-time patches

A transformer treats small regions across several frames as units.

04

Video tokens

A tokenizer converts clips into discrete symbols for sequence prediction.

Six moments in one run

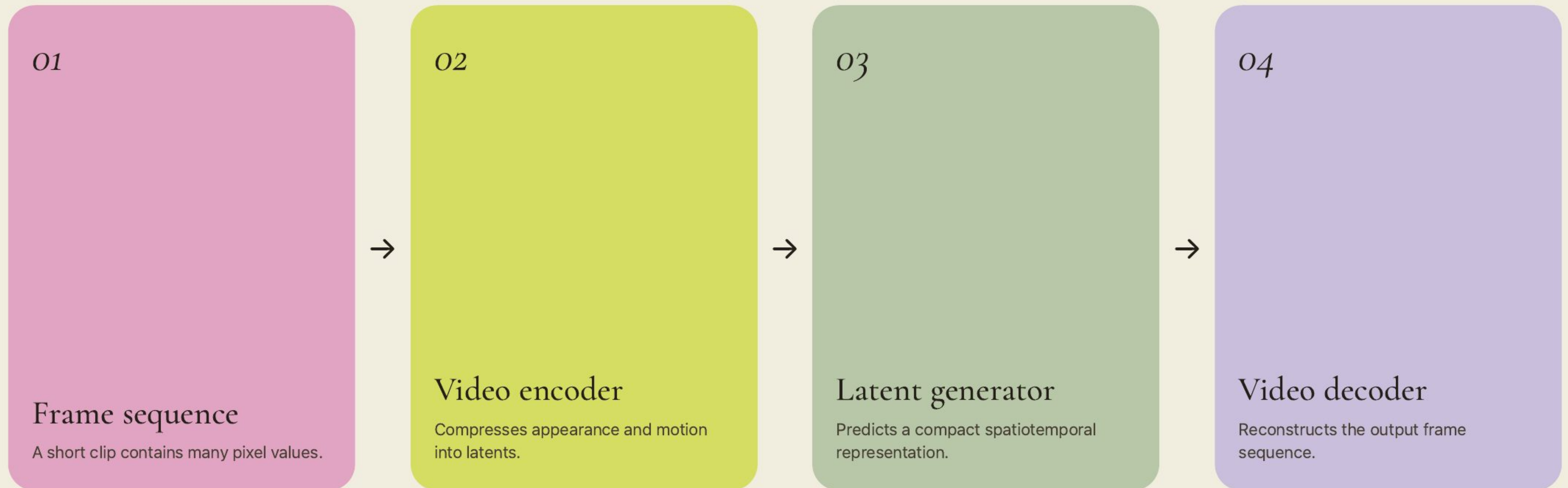
The fox moves because pose and position advance

Each frame is still. Played quickly in order, the sequence becomes a continuous action.



Compression makes video generation practical

The encoder reduces frame size and temporal redundancy, the generator works on compact latents, and the decoder reconstructs visible frames.



The subject moves while identity stays recognizable

A coherent sequence balances these two demands.

Change

Position, pose, expression

These differences create action and performance.

Stability

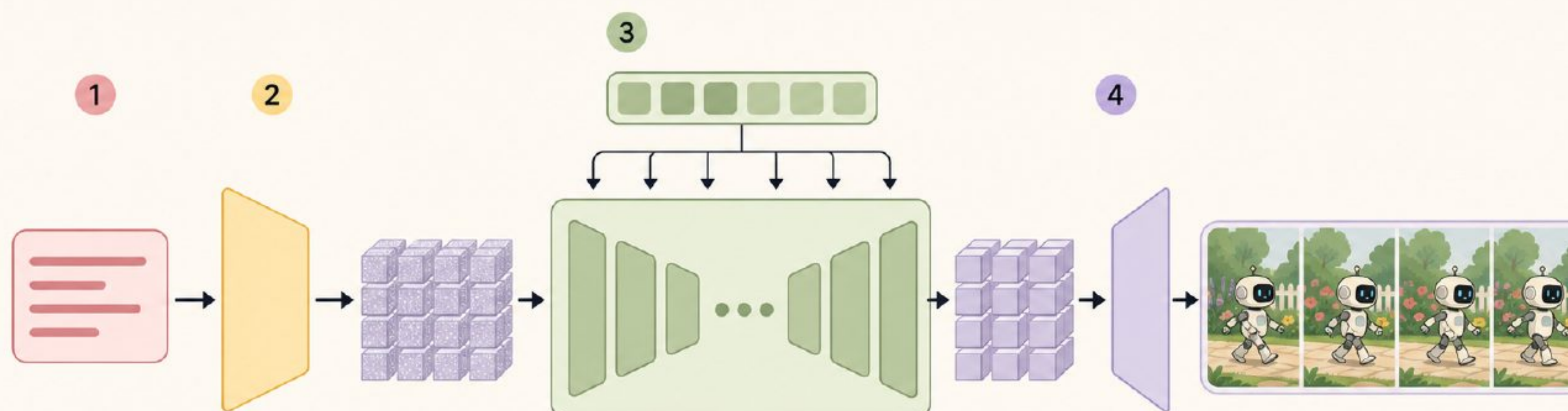
Identity, clothing, object structure

These similarities tell us it is the same subject.

Text encoder + latent video + temporal denoiser + decoder

A modern system coordinates several neural networks

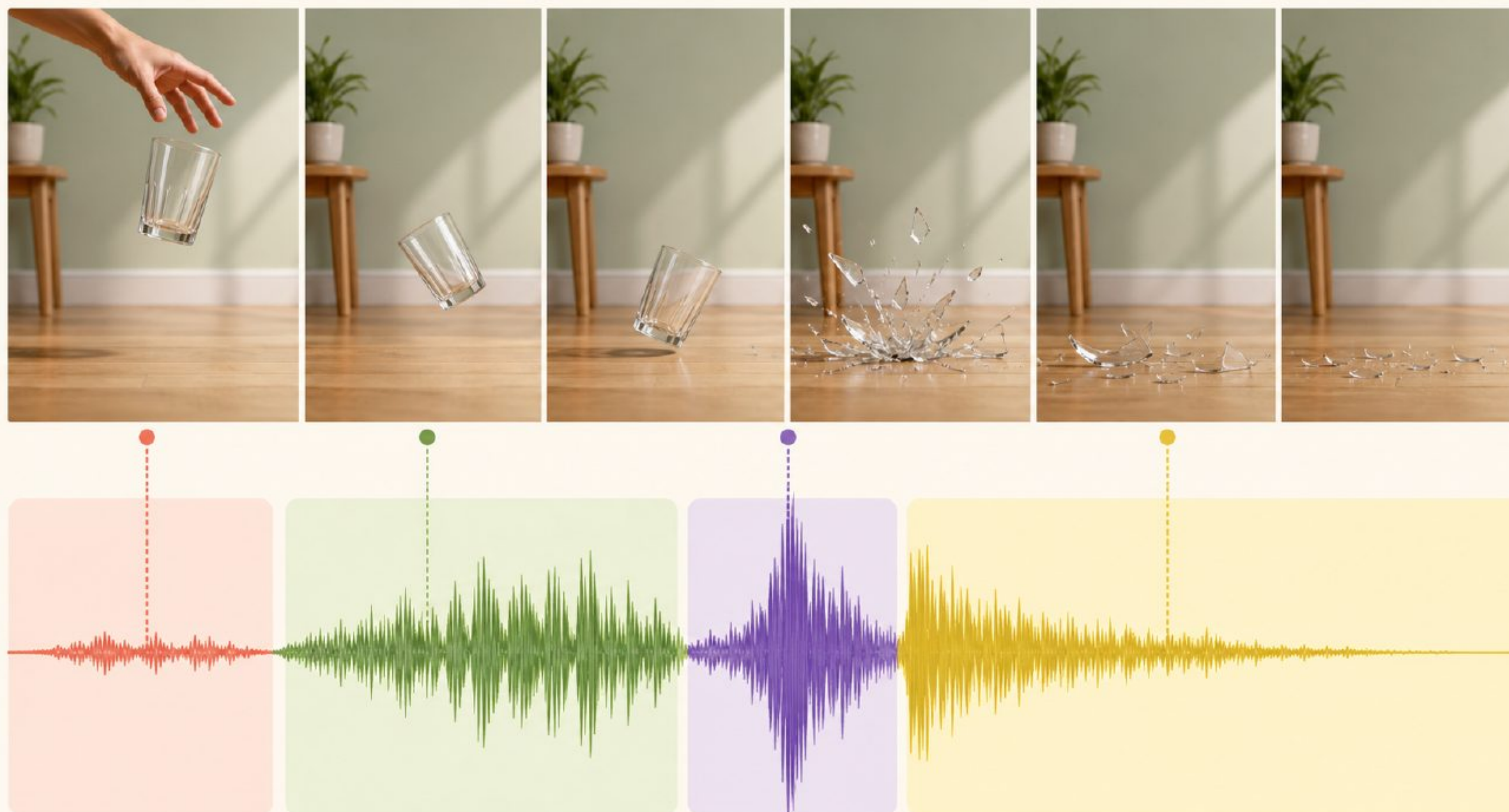
Text embeddings guide a spatiotemporal denoiser, while a video autoencoder moves between compact latent blocks and visible frames.



Align visible events and sound events

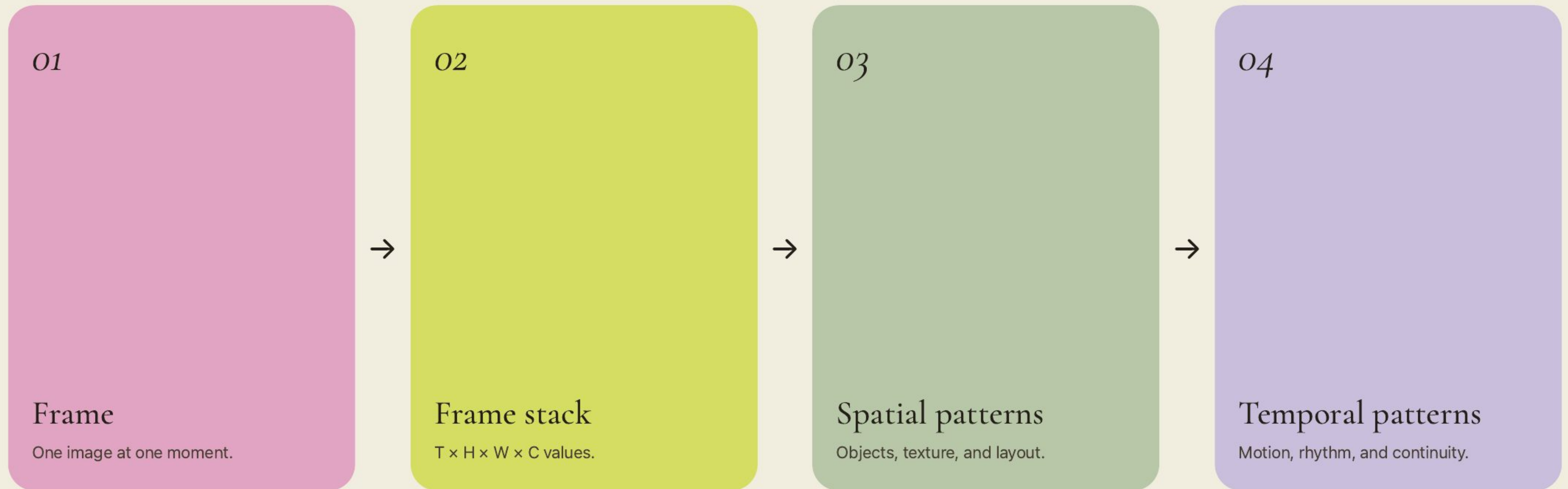
Audio and video must meet at the right moments

The impact frame aligns with the sharp crash. The quieter room echo continues after the glass stops moving.



A video is a stack of image tensors

An image stores height, width, and color channels. Video adds a sequence of time steps, so the model must learn both appearance and change.



Video adds a time dimension to image generation

The model must coordinate appearance, movement, camera, and sound across many frames.

01

Frames

Individual images provide spatial detail.

02

Motion

Changes between frames create action.

03

Continuity

Stable relationships make the sequence believable.

The image-generation idea expands through time

Generating video

A video model can begin with noise and repeatedly predict changes that make an entire sequence more coherent.



Words guide appearance and movement

The prompt describes what happens—not only what exists.

A video prompt needs subjects, actions, environment, camera behavior, pacing, and often an audio plan.

Scene + action + time

One video generator is usually a system of models

The text encoder, video autoencoder, denoiser, and sampler each solve a different part of generation.

01

Text encoder

Converts words into conditioning vectors.

02

Video VAE

Compresses and reconstructs frame sequences.

03

Denoiser

Predicts noise or velocity in latent space.

04

Sampler

Turns many denoising predictions into a trajectory.

Space and time are processed together

Repeated denoising organizes appearance and motion

Each block contains several frames. The airplane, its trajectory, and the sky become coherent across the whole sequence.



Structure arrives before fine detail

Shape, trajectory, and background emerge together

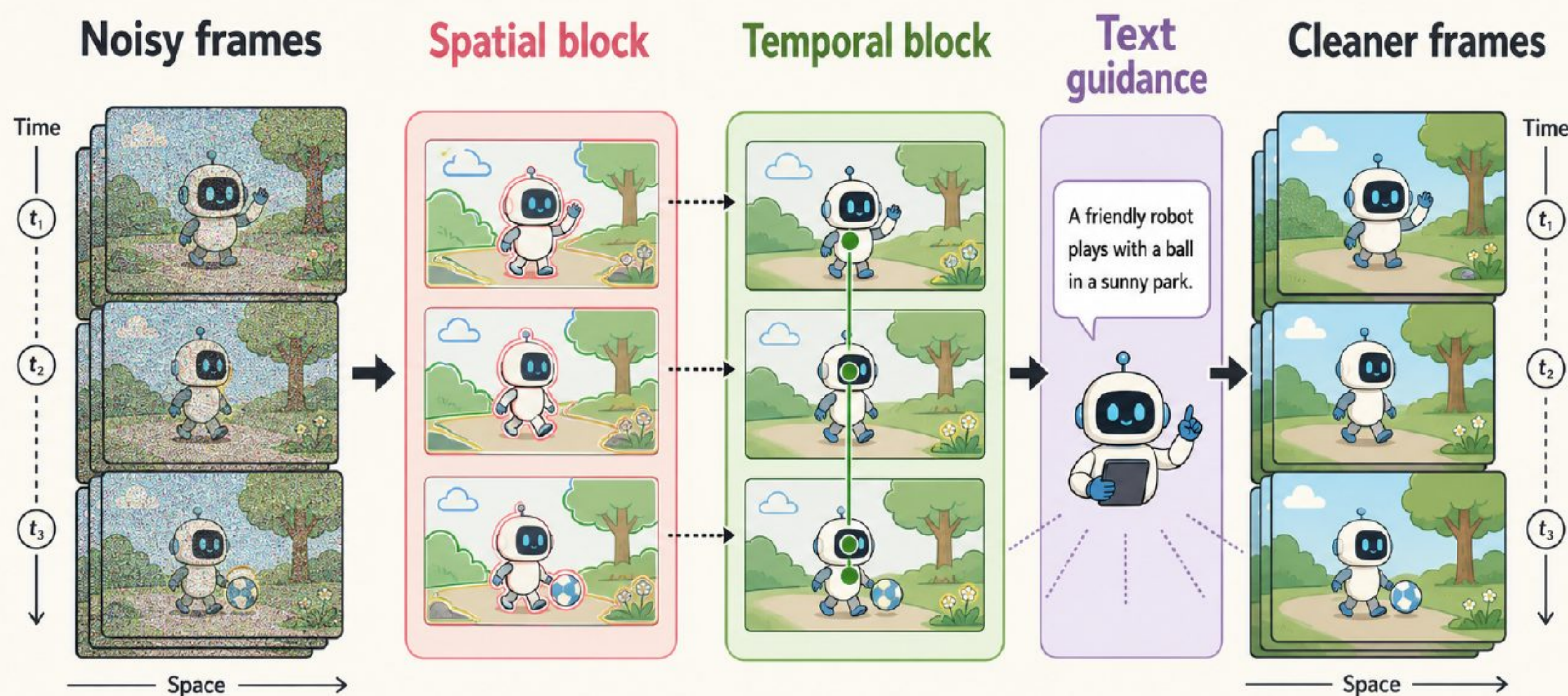
The paper airplane becomes recognizable while its direction and surrounding sky stabilize across frames.



Space answers where · time answers when

One block sees each frame; another connects time

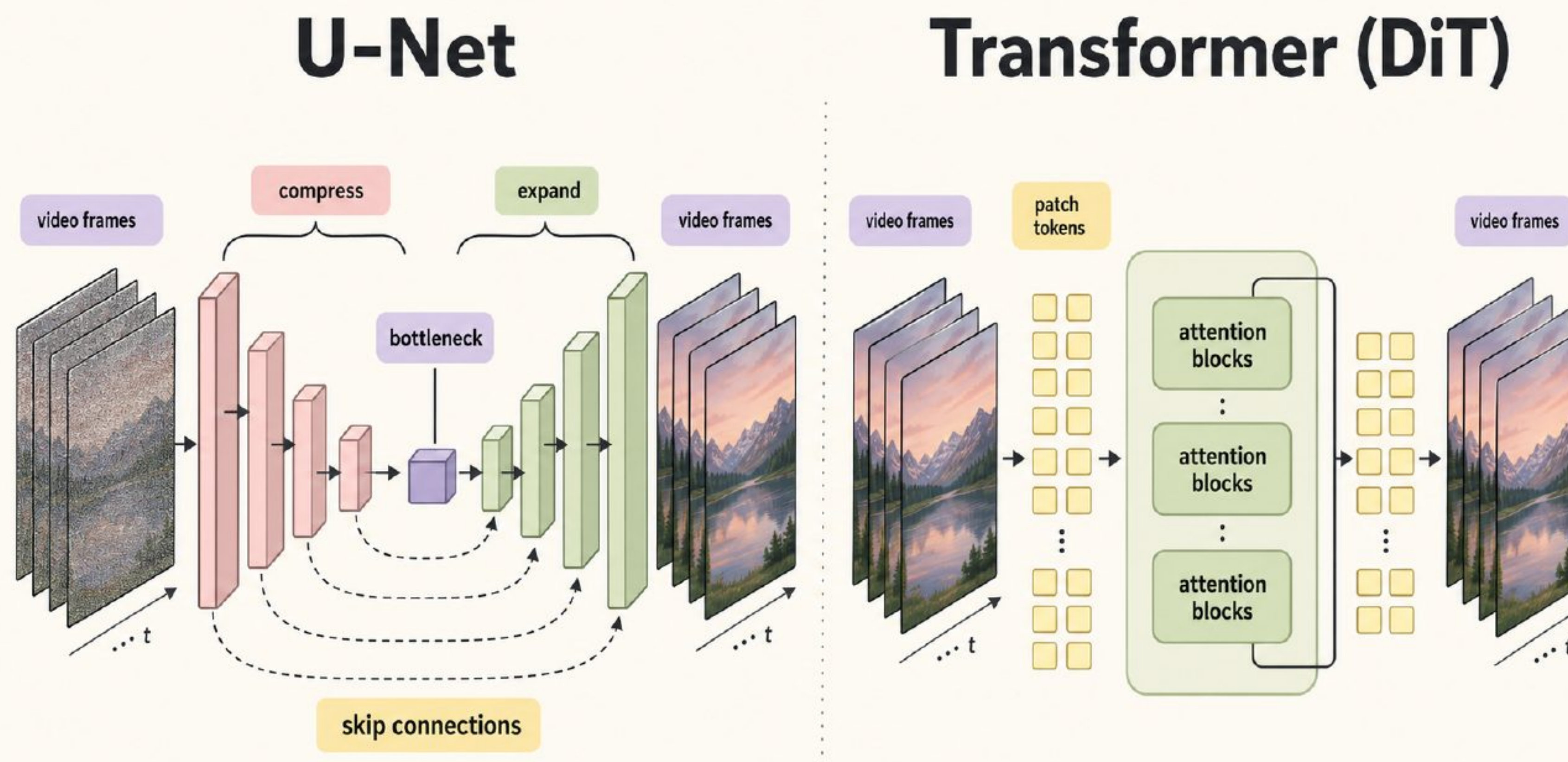
The spatial block finds objects and shapes inside each frame.
The temporal block links the same subject across frames, while
text guidance keeps the result connected to the prompt.



Two structures · the same denoising job

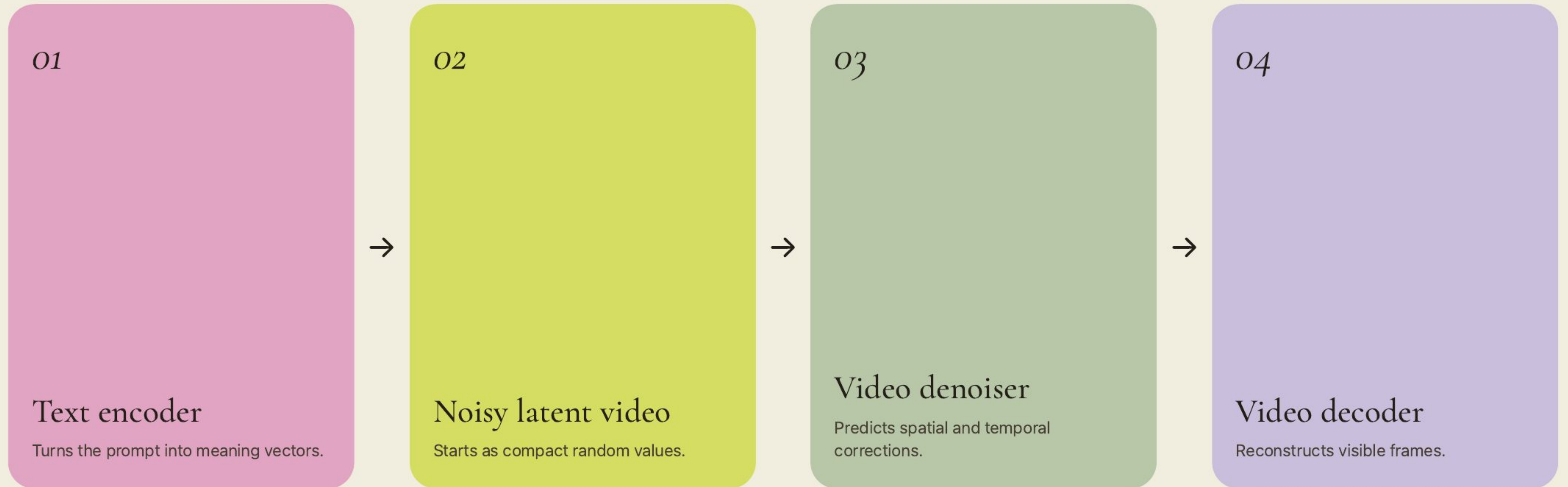
U-Nets use an hourglass; Transformers use patch tokens

A U-Net compresses and expands features while carrying detail through skip connections. A Diffusion Transformer mixes video patch tokens through repeated attention blocks.



Generation usually happens in compressed video space

Text guides a denoising network while a video autoencoder keeps the computation smaller than working with every pixel directly.



Video generation predicts a connected sequence

Text conditions the scene while denoising organizes visual values across space and time.

01

Describe

Define subject, action, setting, and camera.

02

Generate

Build a sequence through repeated predictions.

03

Inspect

Check identity, motion, lighting, and timing.

The hardest errors may appear between frames

Temporal coherence

A coherent video feels like one continuous world rather than a pile of unrelated images.



Relationships survive through time

Temporal coherence means the sequence agrees with itself.

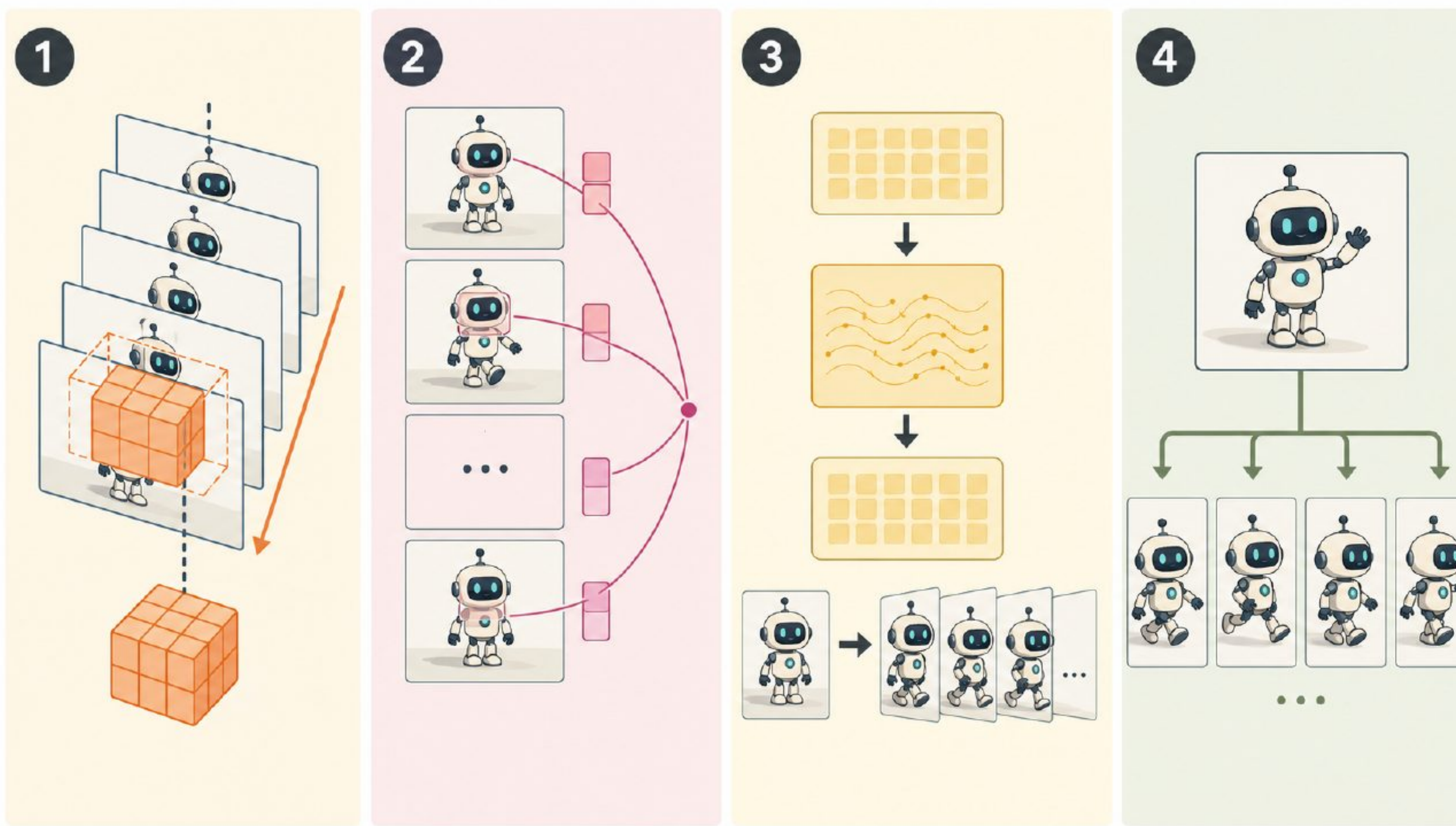
Subjects retain identity, objects follow understandable paths, backgrounds remain stable, and lighting changes for a reason.

Consistency across moments

3D convolution · temporal attention · motion modules · reference conditioning

Four techniques connect frames through time

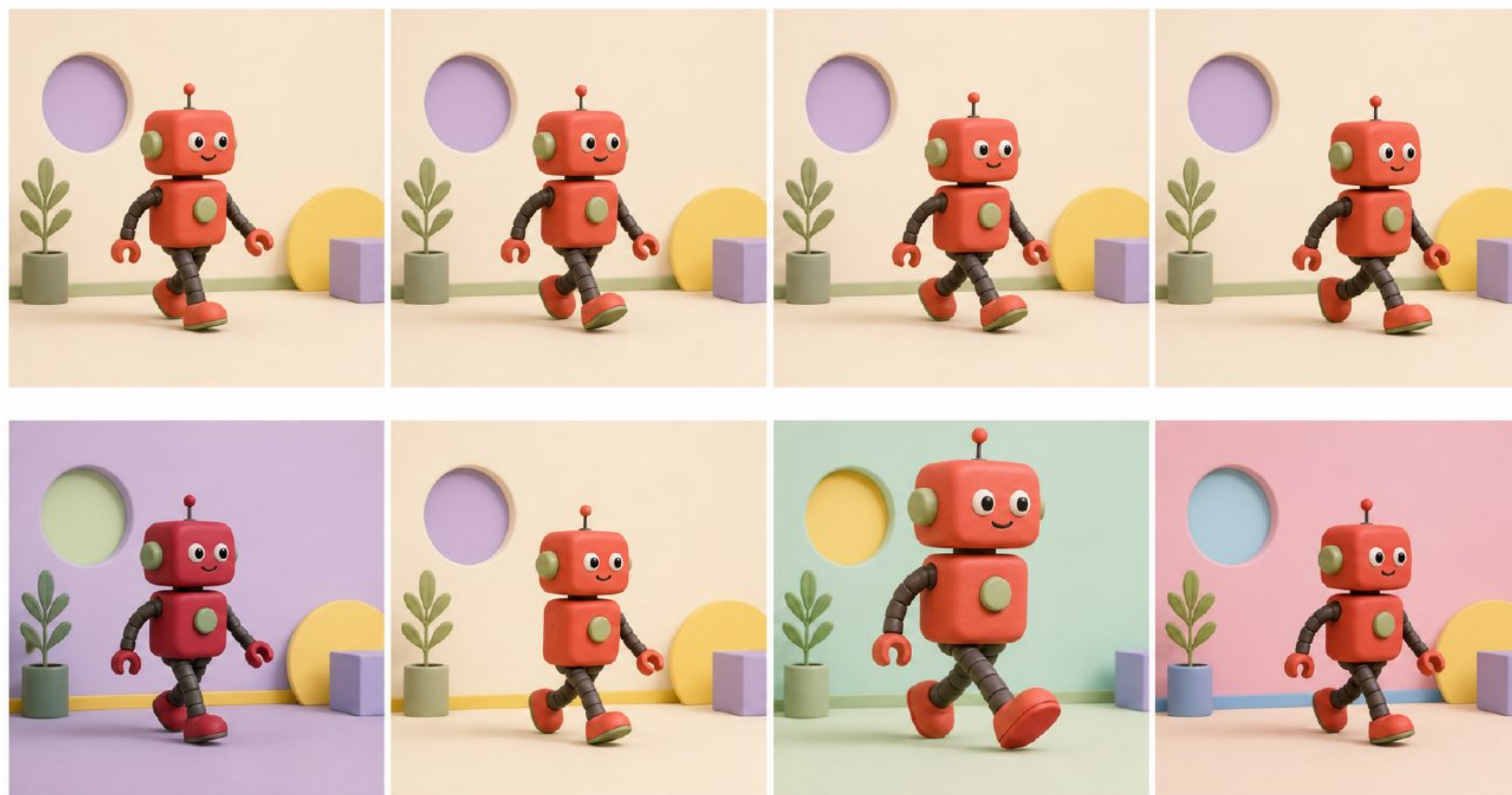
Architectures can share information locally, connect distant moments, insert motion-specific layers, or anchor every frame to one reference image.



Compare two sequences

One robot remains stable; the other quietly changes

Look for missing parts, size jumps, color changes, flickering backgrounds, and impossible motion.



Look for accumulated change

The same character can become a different character

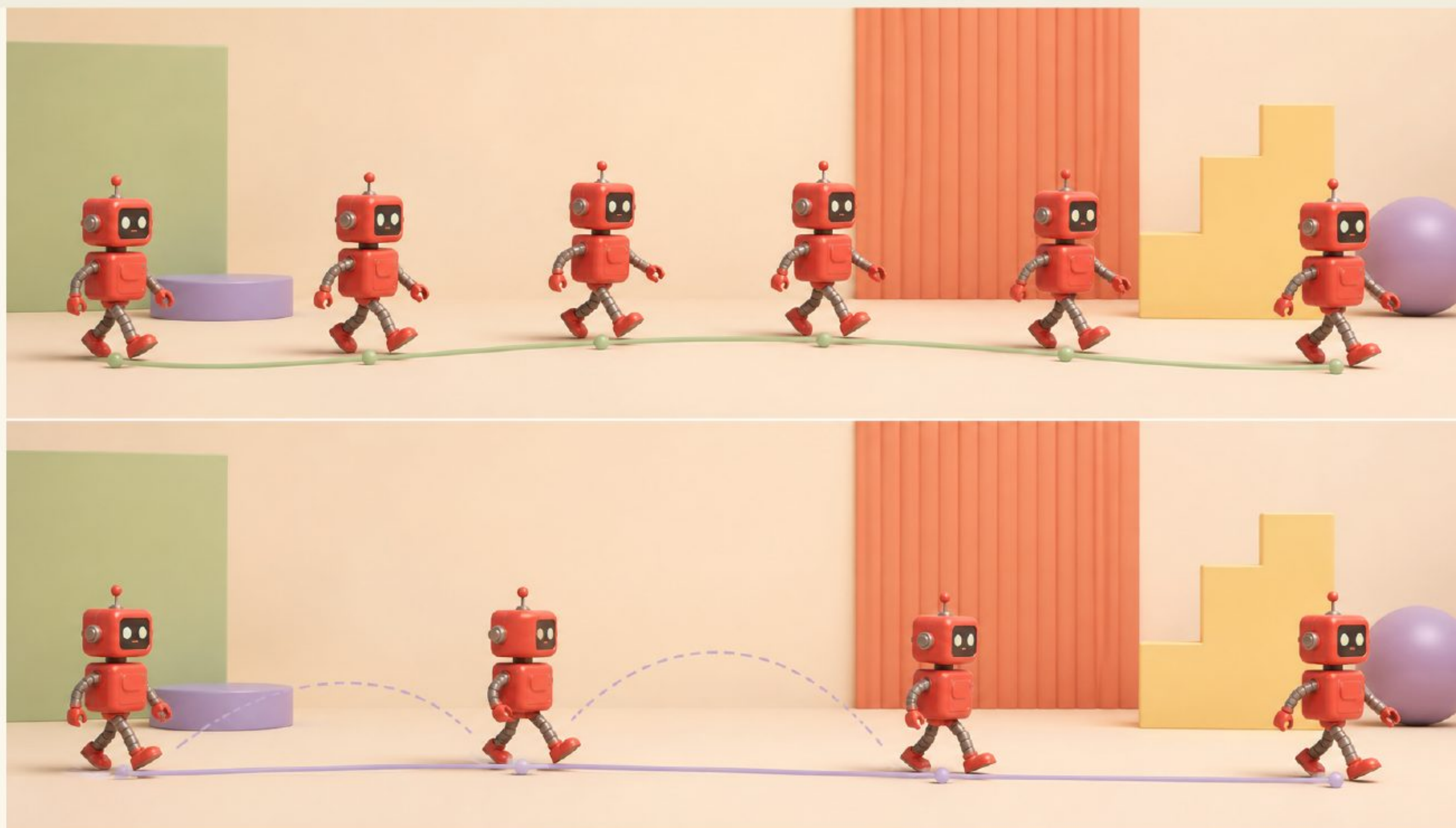
Hair, eye color, clothing details, face proportions, and apparent age drift even though the scene should contain one person.



Intermediate poses create a path

Movement needs a path—not a teleport

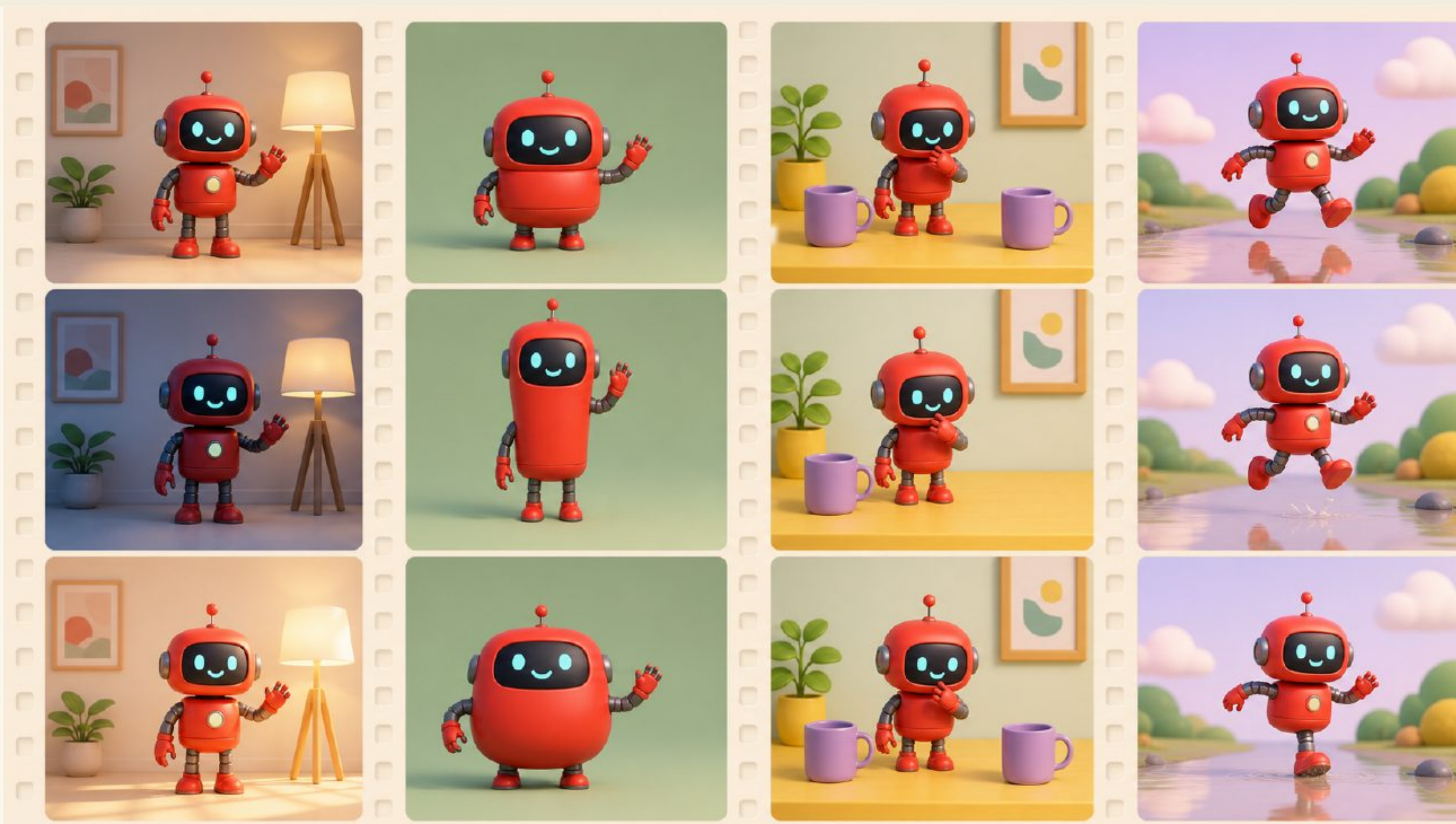
The top sequence passes through continuous walking poses. The bottom sequence jumps between positions and slides unnaturally.



Four ways time can break

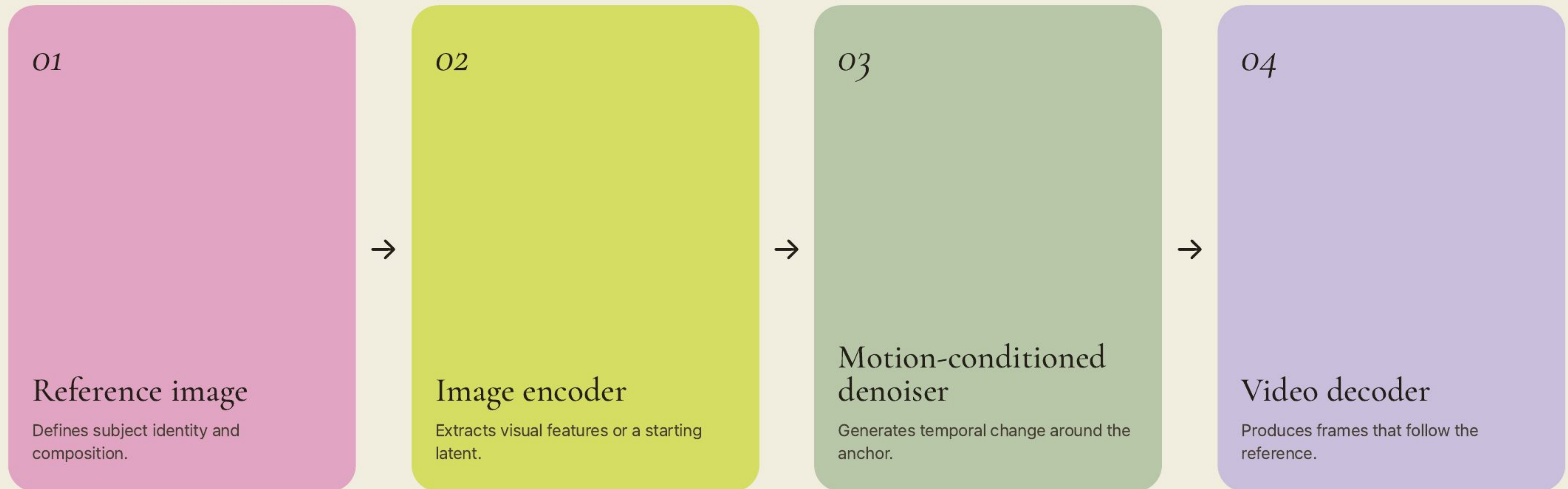
Video errors can hide inside beautiful frames

Look for flickering light, morphing bodies, changing object counts, and impossible contact or reflection physics.



A starting image can anchor every frame

Image-to-video systems encode the reference, reuse its features during denoising, predict motion, and decode a sequence that preserves appearance.



Frames need ways to share information across time

Video models combine several mechanisms so a choice made in one frame can influence nearby or distant frames.

01

3D convolution

Filters extend across width, height, and time to detect short motion patterns.

02

Temporal attention

A frame can compare features with other moments and preserve identity or motion.

03

Motion modules

Extra temporal layers can add video behavior to a model that already understands images.

04

Image conditioning

A starting frame anchors subject appearance, composition, and color.

Turn an idea into a directable scene

Controlling audio & video

A strong prompt coordinates subject, action, environment, camera, style, duration, and sound.



Describe decisions that affect the timeline

Subject + action + setting + camera + light + duration + audio

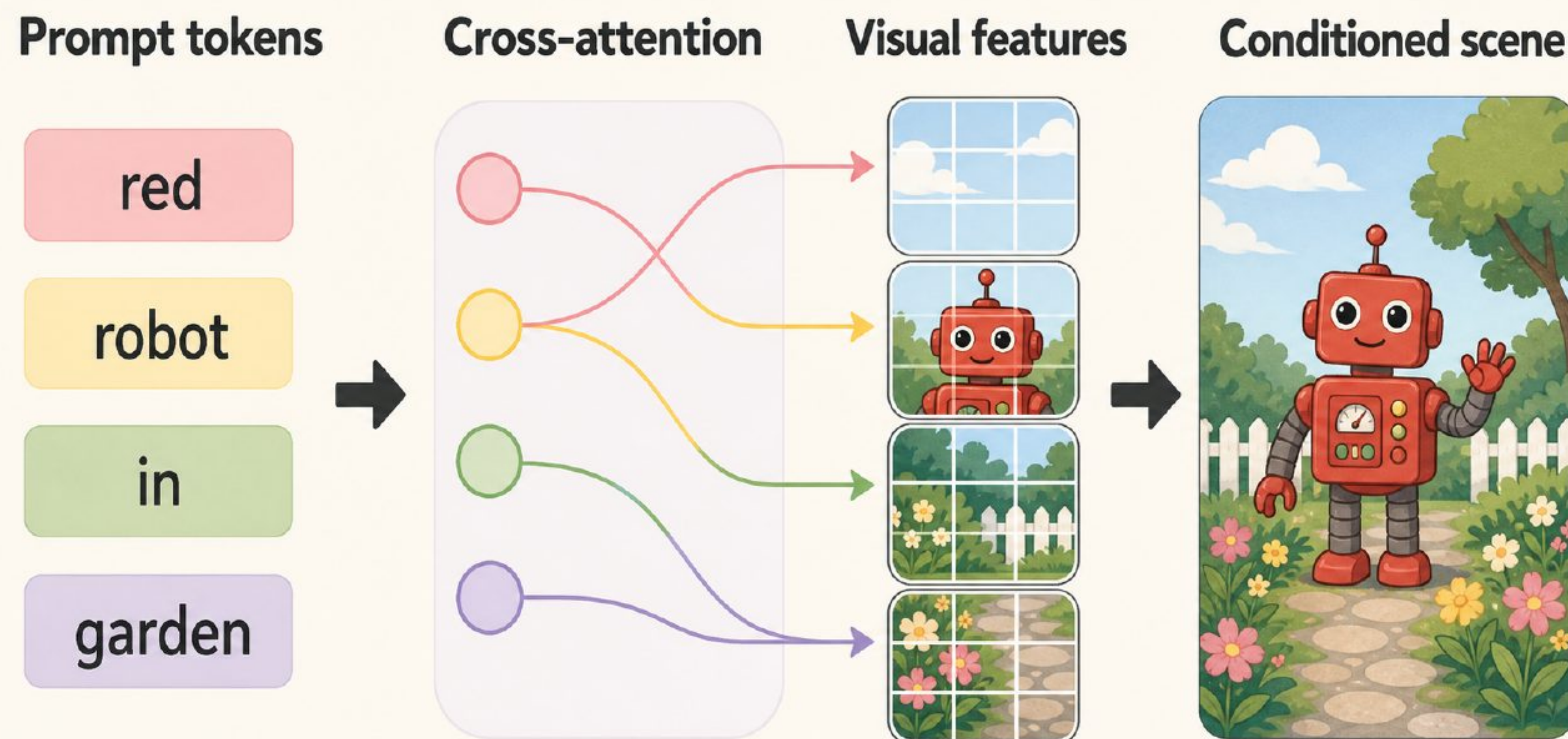
Use the formula as a checklist, then include only the details that clarify the intended scene.

Specific enough to guide · simple enough to follow

Each visual feature looks for useful words

Cross-attention connects prompt words to the scene

Prompt words become tokens. Visual features compare themselves with those tokens and pull in the words that matter—for example, “red” for the robot and “garden” for the background.



Same robot · different attention

Wide, medium, close, and top-down views tell different stories

Camera distance and angle decide whether viewers notice the world, the action, the emotion, or the spatial pattern.



Simple film language creates clearer prompts

Use one camera choice when it helps the audience understand the scene.

01

Static

The camera stays still while action unfolds.

02

Tracking

The camera travels with a moving subject.

03

Pan

The camera turns to reveal or follow.

04

Push in

The camera moves closer to increase focus.

Guidance controls how strongly the model follows the condition

The denoiser combines a conditional prediction with an unconditional prediction. The scale changes the balance between prompt fidelity and natural variation.

Lower guidance

More variation and softer prompt pressure

Motion may look natural, but specific details can drift from the condition.

diverse samples · weaker control

Higher guidance

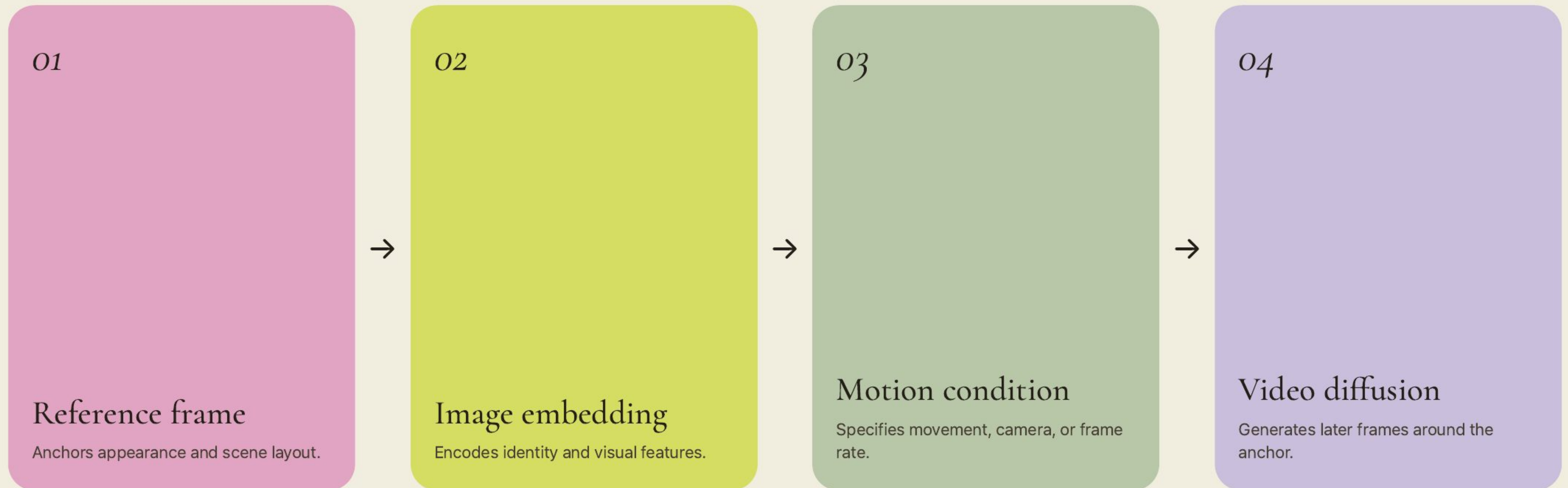
Stronger prompt alignment

Requested details become clearer, but extreme values can create artifacts or reduce diversity.

clearer condition · possible distortion

The first frame provides a strong visual anchor

The reference image supplies identity and composition; motion and camera conditions describe how the latent sequence should evolve.



Modern generators can combine several conditions

Extra inputs reduce ambiguity by constraining identity, geometry, motion, or timing.

01

Image reference

Anchors subject appearance and composition.

02

Depth or edges

Constrains scene geometry and object boundaries.

03

Pose or trajectory

Guides body movement and spatial paths.

04

Audio or timing

Aligns lip motion, beats, or event boundaries.

The input determines how much the model must invent

More visual conditioning usually gives stronger control over identity, composition, and motion.

01

Text-to-video

A text prompt specifies the scene, action, camera, and style; the model invents the visual starting point.

02

Image-to-video

A reference image anchors appearance while the model creates motion and later frames.

03

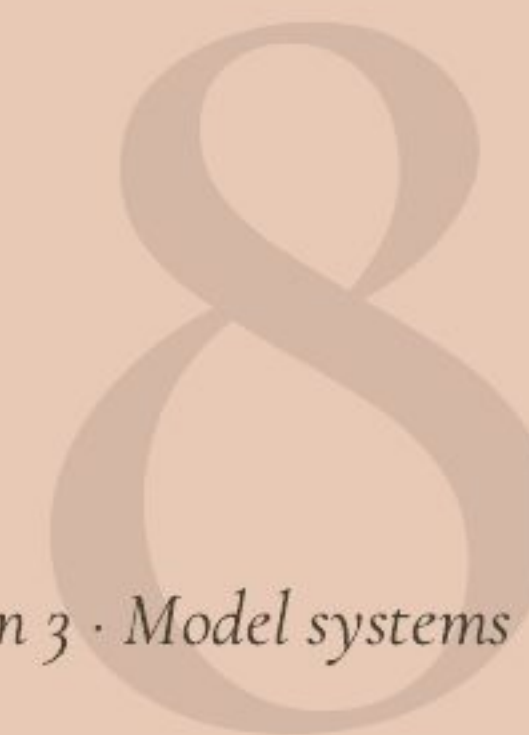
Video-to-video

An input clip supplies timing and motion while generation changes style, subject, or detail.

Several generators can work as one system

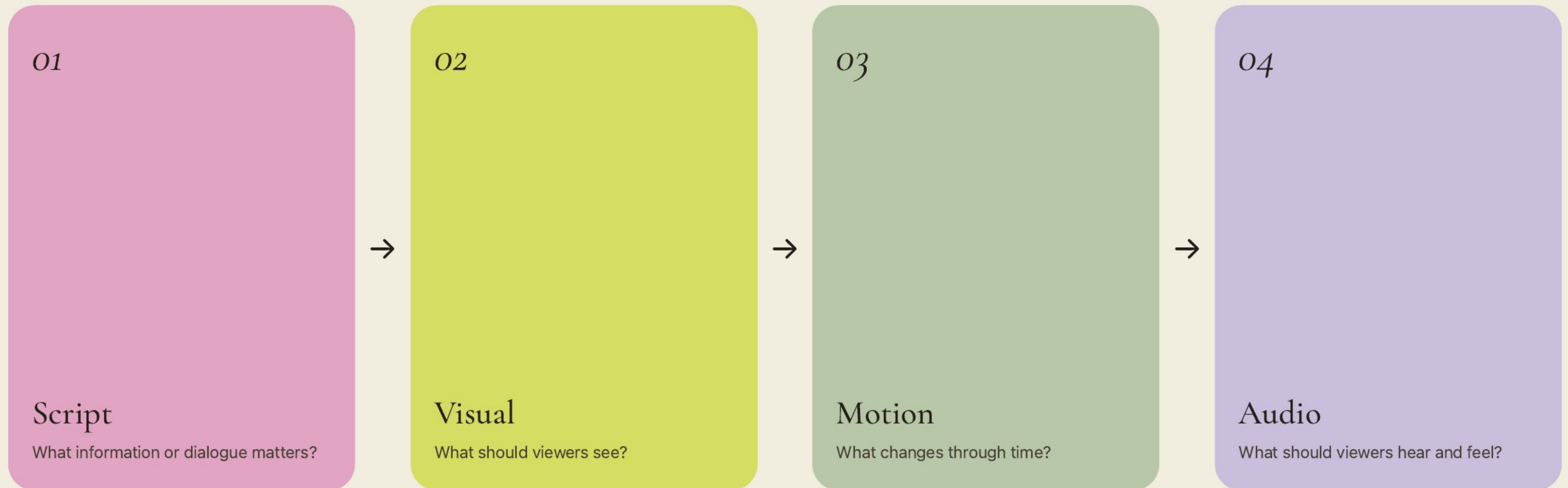
Multimodal generation

A complete media system may connect language, image, video, speech, music, and sound-effect models through shared conditions and a synchronized timeline.



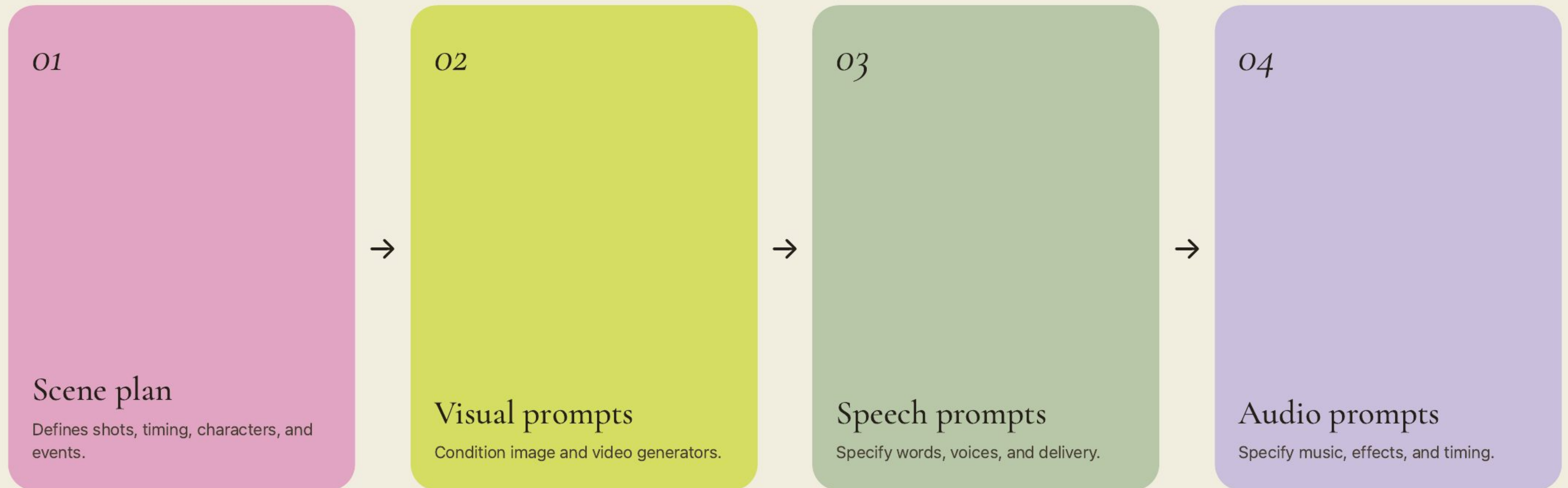
Each layer answers a different question

The layers feel coherent when they support the same purpose and timeline.



One script can become conditions for several generators

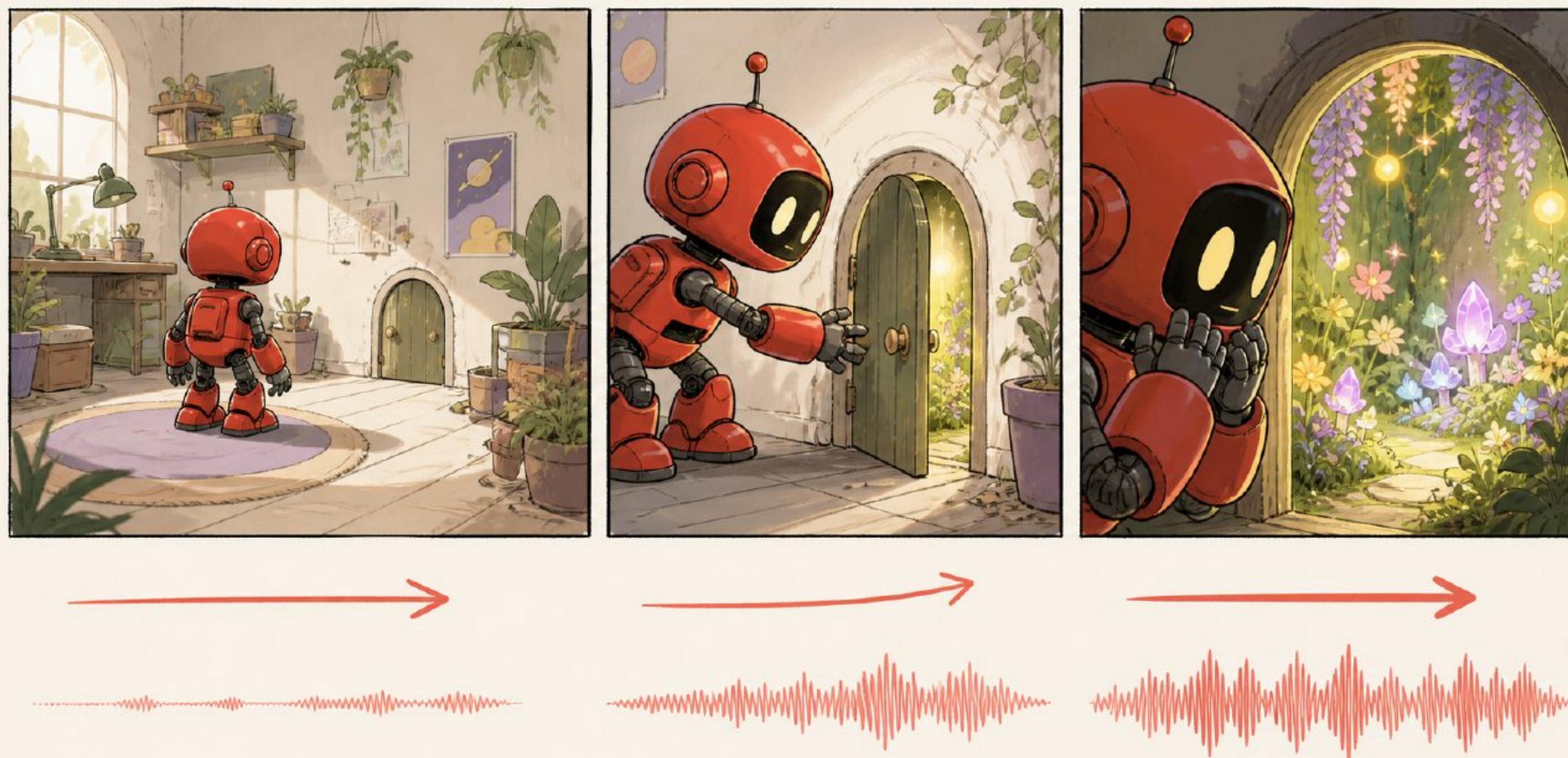
A planning model or human-authored script provides aligned descriptions so separate media models create compatible outputs.



Establish → develop → resolve

A storyboard turns the script into shots

The wide shot introduces the room, the medium shot shows the door opening, and the close-up reveals the discovery.



Generated layers need a shared clock

The system aligns representations so visual events, speech, effects, and music reach the same moments.

01

Timestamps

Place speech words and sound events on the timeline.

02

Frame indices

Connect motion and visual events to exact frames.

03

Beats & phonemes

Provide fine timing for music and lip movement.

04

Shared metadata

Carries character, location, mood, and shot identity.

Media can be generated jointly or assembled as a pipeline

Joint models share representations internally; modular pipelines connect specialized generators through prompts, references, and timing.

Joint model

One network learns several media together

Shared features can improve cross-modal alignment, but training and control are complex.

shared tokens · shared transformer

Modular pipeline

Specialized models exchange conditions

Components are easier to replace and inspect, but outputs must be synchronized carefully.

TTS + video + music + renderer

Several research systems shaped modern video generation

These examples are not identical products; they show how the field developed ideas for motion, scale, text guidance, and latent video.

01

Video Diffusion Models

Extended image diffusion into time and demonstrated joint modeling of frames.

02

Make-A-Video

Combined image-text knowledge with unlabeled video to learn what scenes look like and how they move.

03

Imagen Video

Used a cascade of diffusion models to increase temporal and spatial resolution.

04

Stable Video Diffusion

Scaled latent video diffusion and supports image-conditioned video generation.

A story is a system of timed relationships

Multimodal creation works when each medium contributes something useful rather than repeating everything.

01

Plan

Give the scene one central change.

02

Coordinate

Align script, shots, motion, and sound.

03

Focus

Remove layers that do not serve the story.

Models learn first, then generate

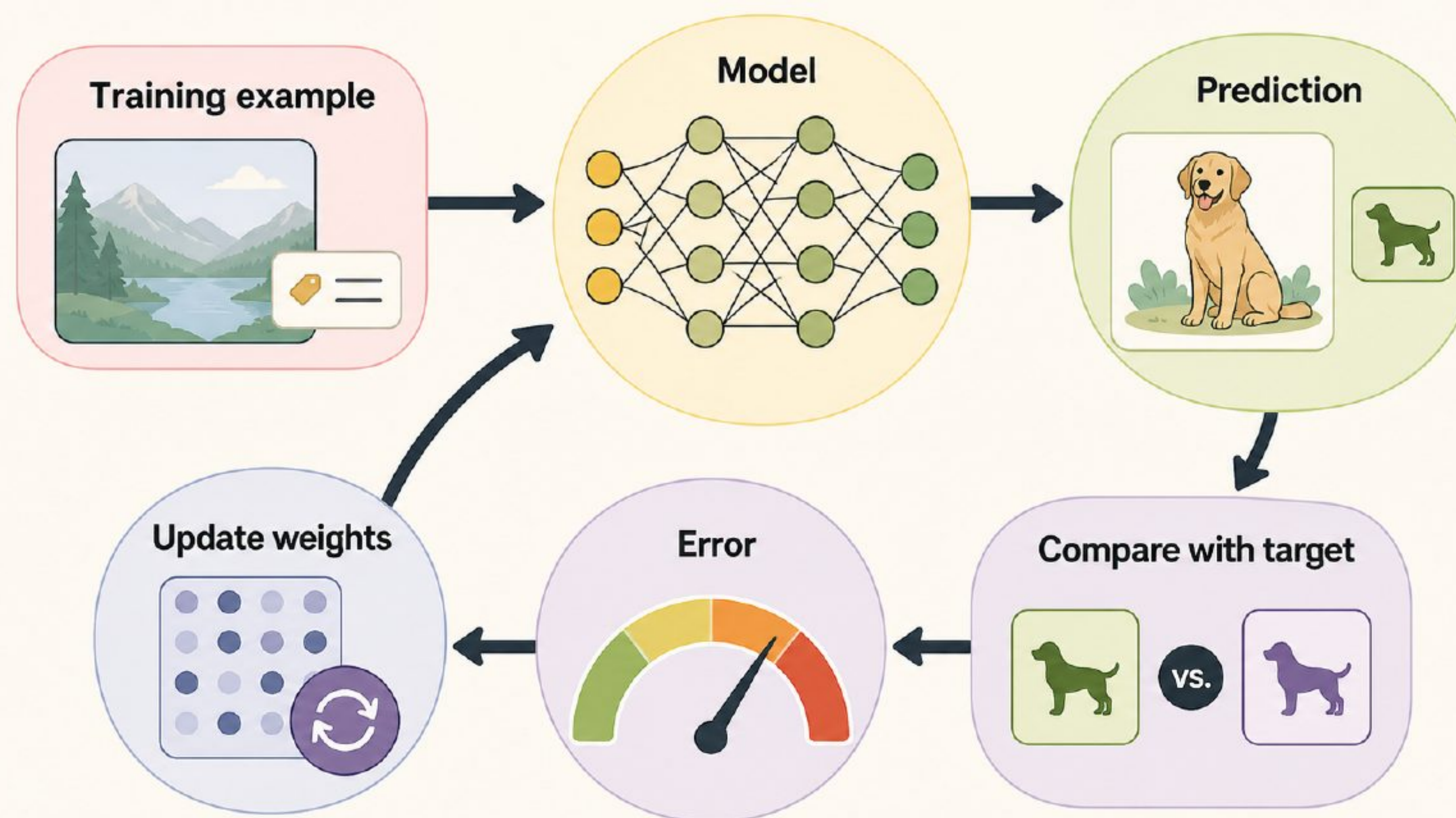
Training, inference & evaluation

Training teaches a network to predict missing structure. Inference applies that learned prediction repeatedly, and evaluation measures quality, control, coherence, and speed.

Predict · compare · update · repeat

Training changes the model after every measured error

The model makes a prediction, compares it with the target, and measures the difference. That error sends an update back into the model so the next prediction can improve.



Different supervision teaches different abilities

The model architecture and objective determine which datasets are useful and what controls the final system can understand.

01

Paired text–media

Captions teach relationships between language and audio or video.

02

Unlabeled media

Large audio or video collections teach natural structure and motion.

03

Reference pairs

Related clips or voices teach transformation and conditioning.

04

Curated examples

Filtering improves relevance, quality, and representation.

Each model learns a different prediction game

The loss function tells the network which error to reduce during training.

01

Noise prediction

Diffusion learns the noise or velocity needed to reverse corruption.

02

Next-token prediction

Autoregressive models learn the next codec or video token.

03

Reconstruction

Autoencoders learn to compress media while preserving important detail.

04

Condition alignment

Text and reference signals must influence the correct content.

A model can improve one metric and weaken another

Useful evaluation combines automated measures with listening, viewing, and task-specific human judgments.

01

Fidelity

Does the output sound or look realistic and detailed?

02

Condition match

Does it follow the text, reference, timing, or motion control?

03

Temporal coherence

Do identity, motion, and audio remain consistent through time?

04

Efficiency

How much memory, computation, and waiting time are required?

Learning weights and generating media are different processes

Training is expensive and repeated across huge datasets. Inference keeps the weights fixed and uses them to produce one requested output.

Training

Predict, measure error,
update weights

Many examples teach the model general patterns and conditioning relationships.

data → loss → gradient → new weights

Inference

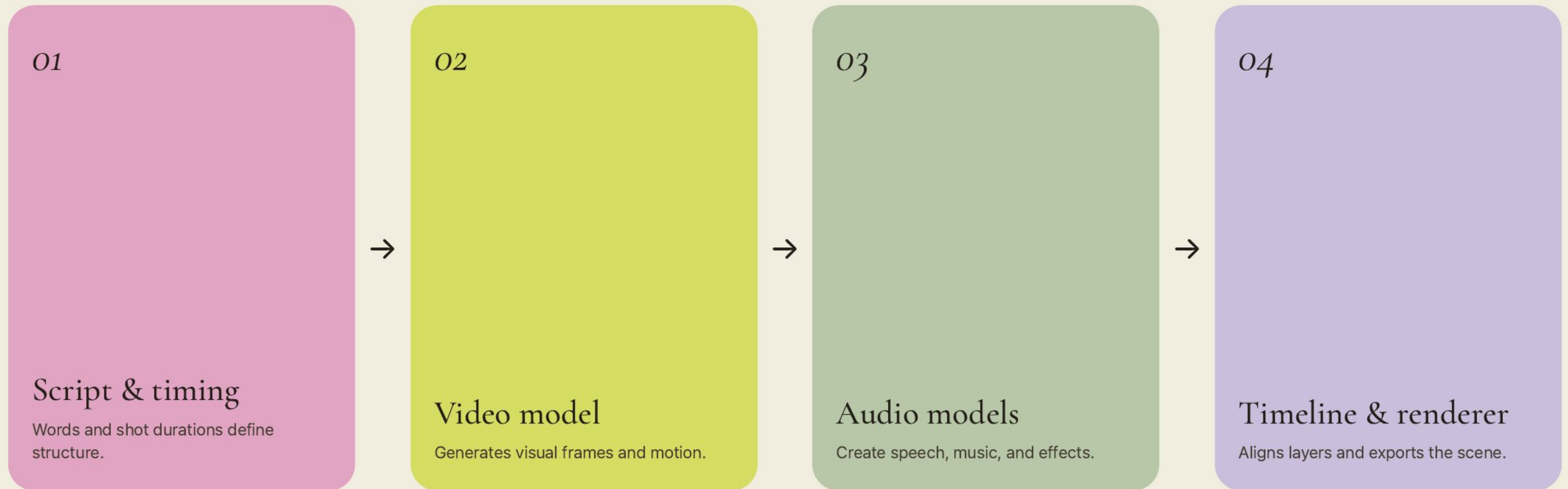
Start from context or
noise, then sample

Prompts, references, seeds, and guidance control one generation run.

condition → predictions → media

A short scene may use several specialized models

The components can be generated separately, aligned on a shared timeline, and rendered into one experience.



Small settings can change the generated result

These controls affect speed, prompt strength, diversity, and the amount of temporal detail without retraining the model.

01

Seed

Chooses the initial noise pattern. Reusing it can make comparisons more controlled.

02

Sampling steps

More denoising steps may improve refinement, but they increase generation time.

03

Guidance

Higher guidance follows the condition more strongly, but extreme values can reduce natural variation.

04

Frames & resolution

Longer, faster, or larger videos require more memory and more computation.

Different media need different generation engines

Model architectures & practical limits

Speech, music, sound effects, and video can use diffusion, autoregressive prediction, neural codecs, transformers, or combinations of several techniques.

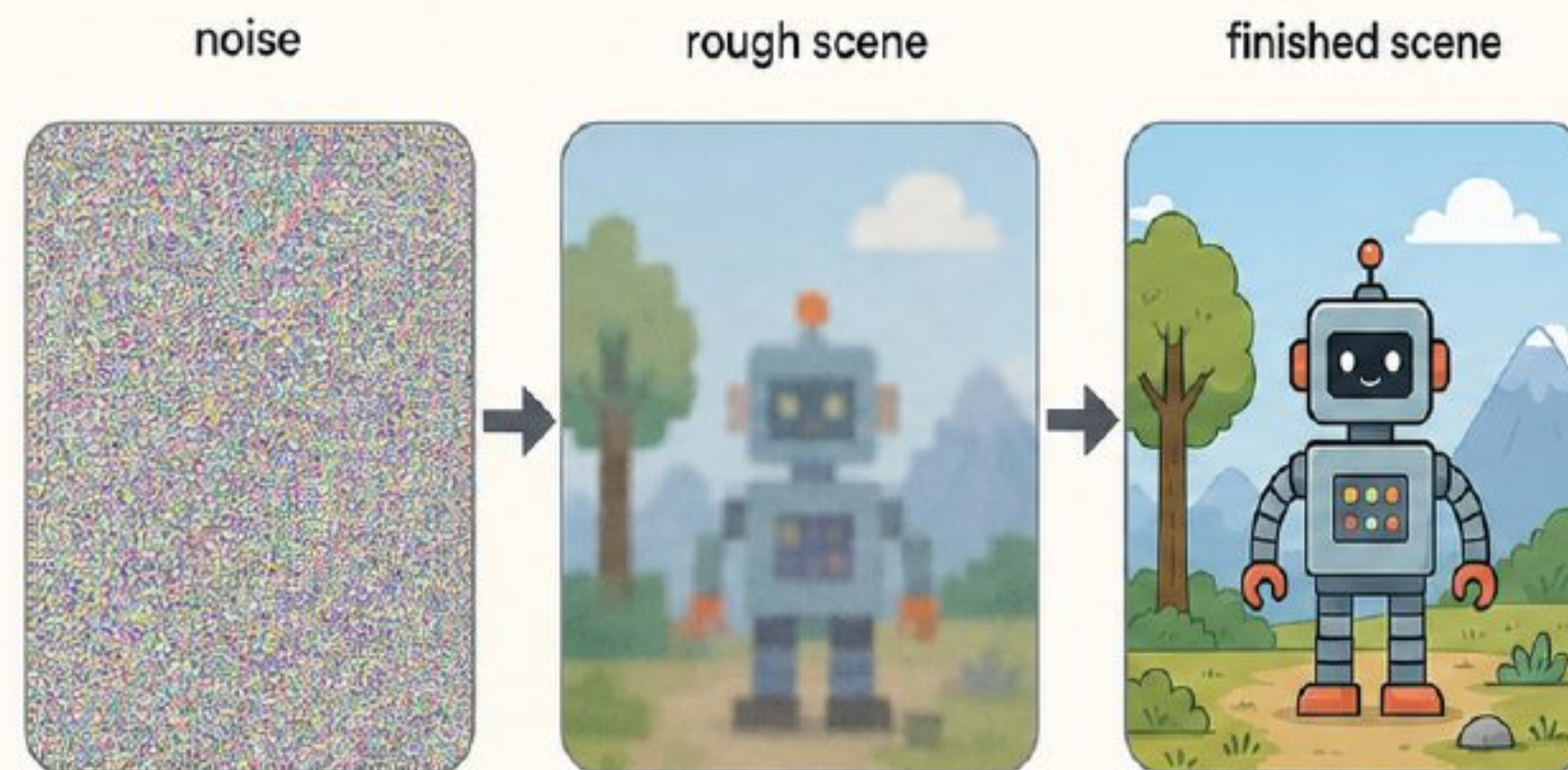
Refine everything · or extend a sequence

Diffusion and autoregressive models build media differently

Diffusion improves a whole noisy structure over several steps.
An autoregressive model predicts the next token using the tokens that came before it.

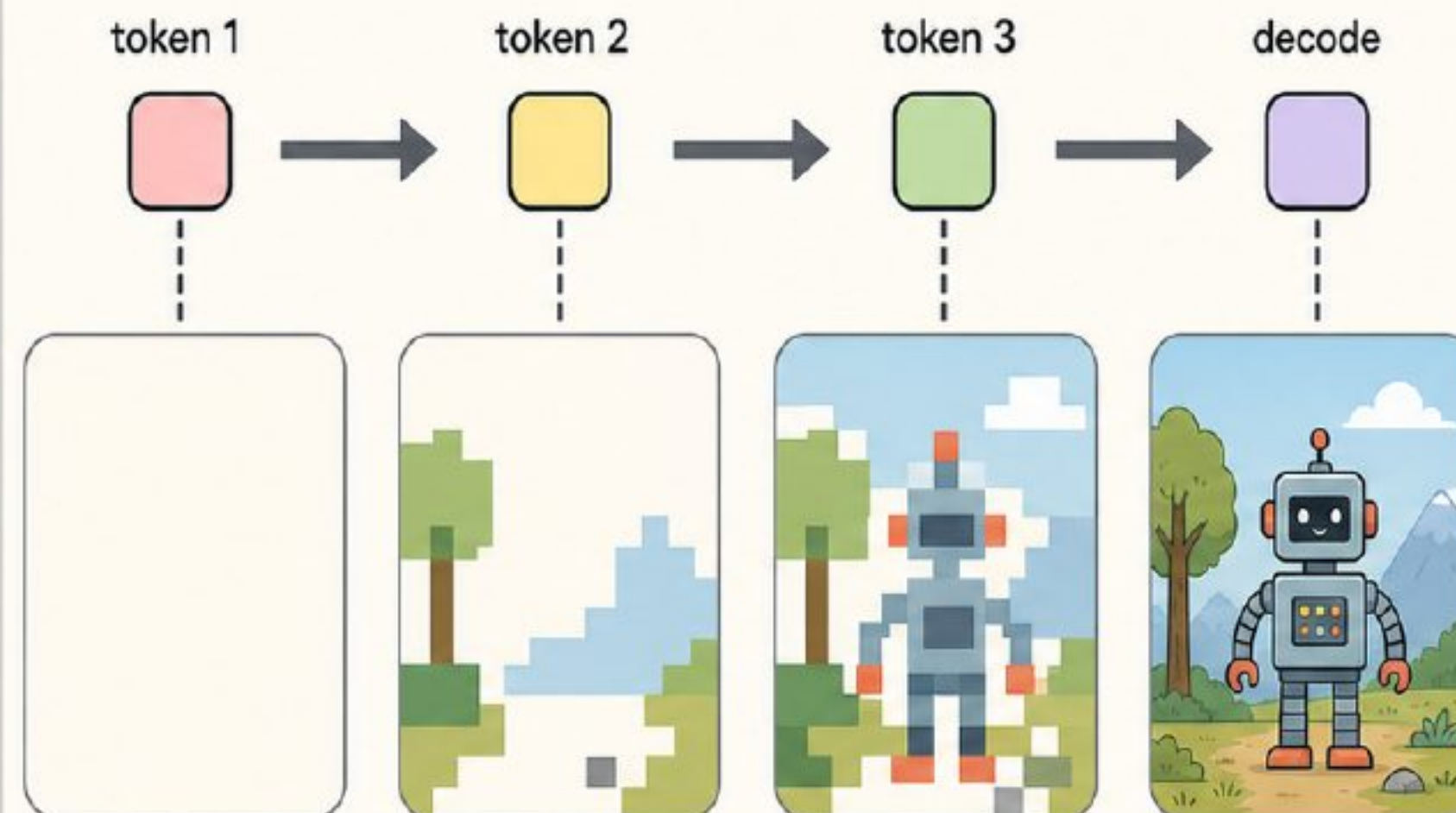
Diffusion

Refine the whole structure



Autoregressive

Predict the next token



Match each speech model to its main prediction idea

This recap connects model names to the visual architectures introduced earlier. Focus on what each system predicts in the middle of the pipeline.

01

Tacotron 2

Predicts mel-spectrogram frames one after another, then uses a vocoder to make sound.

02

FastSpeech 2

Predicts timing, pitch, energy, and speech frames in parallel.

03

VALL-E

Predicts discrete speech tokens using a language-model-like Transformer.

04

NaturalSpeech 2

Uses diffusion to refine a compact latent representation of speech or singing.

Match each audio model to the representation it predicts

The model name matters less than the prediction target: samples, token layers, music tokens, or a compressed latent.

01

WaveNet

Predicts the next raw waveform sample from earlier samples.

02

AudioLM

Predicts meaning tokens first, then coarse and fine audio tokens.

03

MusicLM

Predicts a musical plan and then detailed acoustic token streams.

04

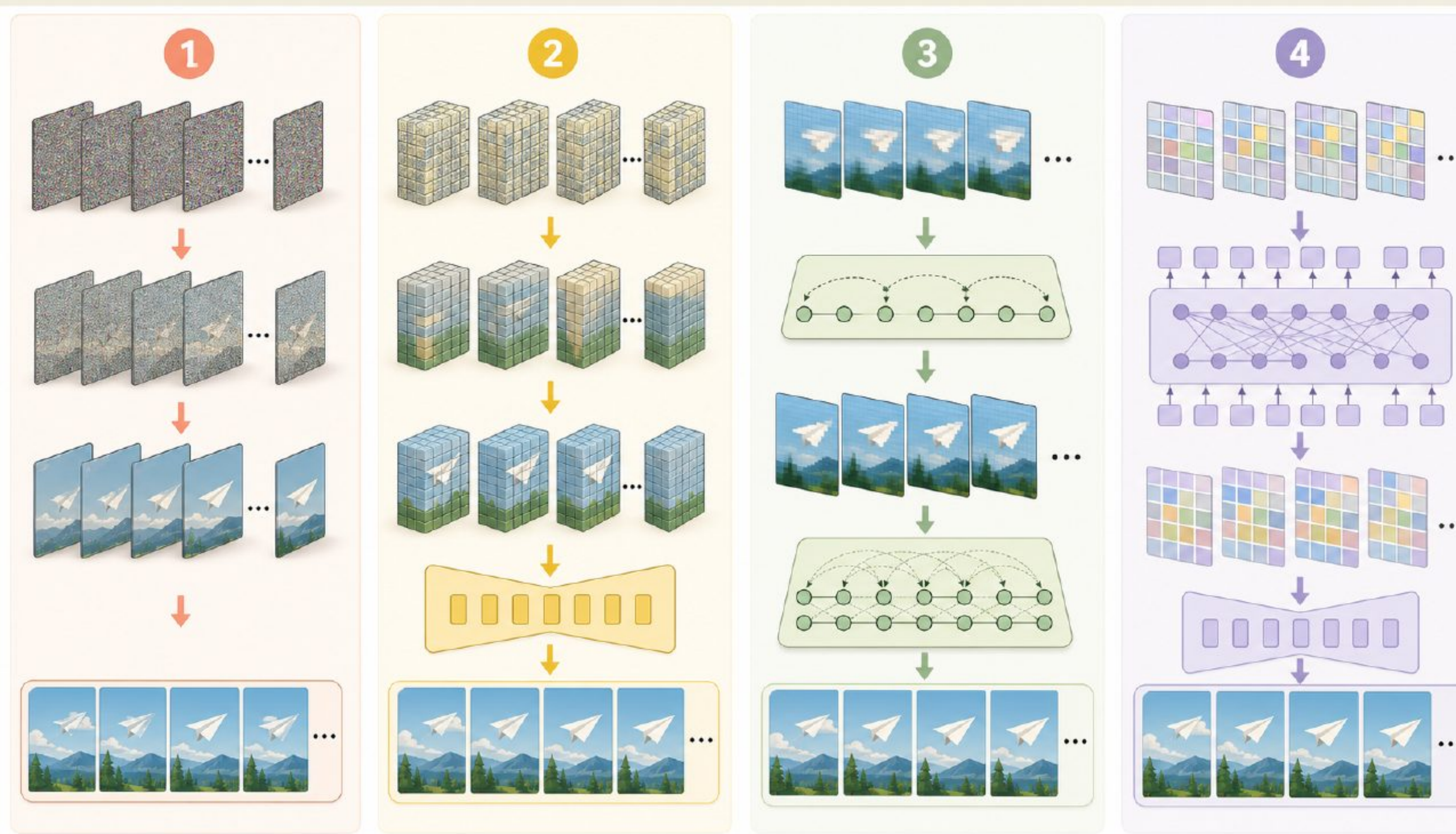
AudioLDM

Starts with a noisy audio latent and refines it under text guidance.

Pixels · latents · cascades · tokens

Four structures can generate the same moving scene

Pixel diffusion denoises frames directly, latent diffusion works in compressed space, cascades add resolution in stages, and token transformers predict discrete video units.



No model wins every dimension at once

Understanding the trade-offs helps explain why short clips and carefully constrained generations are often more reliable.

01

Quality vs. speed

More computation can improve detail, but generation becomes slower and more expensive.

02

Duration vs. consistency

Longer sequences give more opportunities for identity, motion, or scene structure to drift.

03

Control vs. diversity

Strong conditioning narrows the result; weaker conditioning leaves more room for variation.

04

Compression vs. detail

Latent and codec representations save computation, but must preserve the details people notice.

Can you explain the complete media pipeline?

Use one short scene to connect representation, conditioning, generation models, temporal coordination, and revision.

01

Represent

Sound becomes samples or tokens;
video becomes frames or latents.

02

Condition

Prompts and references guide voice,
action, camera, style, and audio.

03

Generate

Diffusion or autoregressive models
predict the media representation.

04

Refine

People compare outputs, tune
settings, coordinate layers, and revise.

Different models contribute different strengths

Sound and motion become stories through coordinated generation.

Speech, music, effects, images, and motion may come from different model families. The final experience depends on how their outputs are directed, aligned, and revised.

Represent → condition → generate → coordinate → refine



A short scene becomes coherent when every generated layer supports one idea.